ME101: (3 1 0 8)

PLANAR KINETICS OF RIGID BODIES

LECTURE: 34-37

Plane Kinetics of Rigid Bodies

:: **Relates external forces** acting on a body with the **translational** and **rotational motions** of the body

:: Discussion restricted to **motion** in a **single plane** (for this course)

- Body treated as a **thin slab** whose **motion** is **confined to the plane of slab**
  - **Plane** containing **mass center** is generally considered as **plane of motion**
  - All **forces** that act on the body get **projected** on to the **plane of motion**
- **All parts of the body** move in **parallel planes**
- A **body** with **significant dimensions normal** to the **plane of motion** may be treated as having **plane motion** if the **body is symmetrical** about the **plane of motion** through the **mass center**
  - Idealizations suitable for a very large category of **rigid body motions**

Plane Kinetics of Rigid Bodies

:: Earlier discussion on **rectilinear/curvilinear motion**

- 2 equations of motion

$$\sum F_x = ma_x$$
$$\sum F_y = ma_y$$

:: Plane kinetics of rigid bodies

- Additional equation of motion

- Account for the **rigid body rotation**

Rigid Body Kinetics :: Force/Mass/Acc

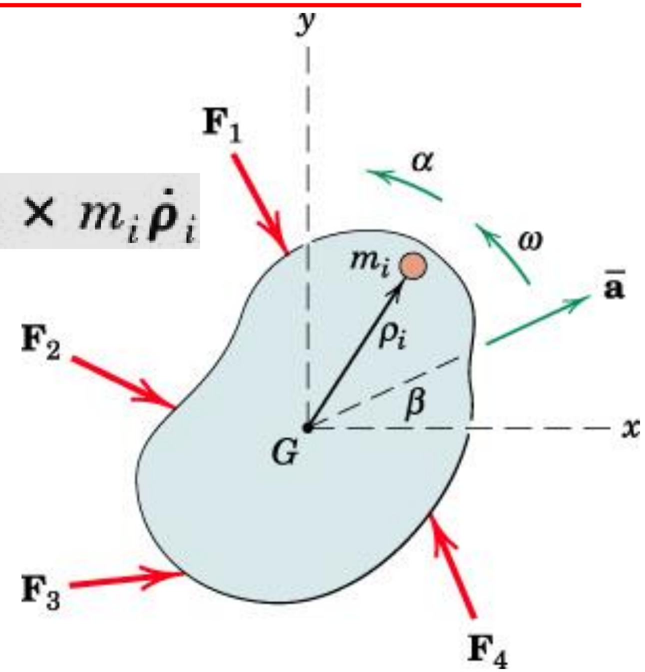
Plane Motion Equations

$\boldsymbol{\omega} = \omega \mathbf{k}$; $\boldsymbol{\alpha} = \alpha \mathbf{k}$; $\alpha = \dot{\omega}$

Angular momentum @ *G (Mass Center)* $\mathbf{H}_G = \Sigma \boldsymbol{\rho}_i \times m_i \dot{\boldsymbol{\rho}}_i$

Vel of m_i relative to G $\dot{\boldsymbol{\rho}}_i = \boldsymbol{\omega} \times \boldsymbol{\rho}_i$

$\boldsymbol{\rho}_i \times \dot{\boldsymbol{\rho}}_i$ is a vector normal to the *x-y* plane along ω (magnitude = $\rho_i^2\omega$)



Magnitude of $\mathbf{H}_G$: $H_G = \Sigma \rho_i^2 m_i \omega = \omega \Sigma \rho_i^2 m_i$

The summation: $\Sigma \rho_i^2 m_i = \int \rho^2 \, dm$

→Mass Moment of Inertia ($\bar{I}$) of the body about *z*-axis through *G*

→Measure of the rotational inertia, which is the resistance to change in rotational velocity due to the radial distribution of mass around the *z*-axis through *G*

$H_G = \bar{I}\omega$ → $\Sigma M_G = \dot{H}_G = \bar{I}\dot{\omega} = \bar{I}\alpha$

→ Generalized laws of motion:

$$\boxed{\begin{aligned} \Sigma \mathbf{F} &= m\bar{\mathbf{a}} \\ \Sigma M_G &= \bar{I}\alpha \end{aligned}}$$

Rigid Body Kinetics :: Force/Mass/Acc

Moment Equations

Moment @ any arbitrary point *P*:

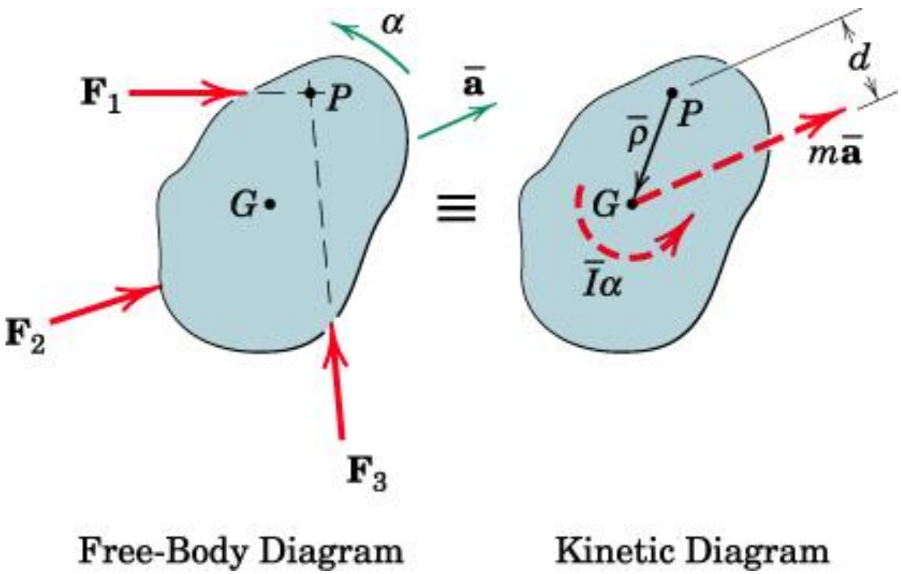
$$\Sigma \mathbf{M}_P = \dot{\mathbf{H}}_G + \bar{\boldsymbol{\rho}} \times m\bar{\mathbf{a}}$$

$\bar{\boldsymbol{\rho}}$ is the vector from *P* to mass center *G*, and $\bar{\mathbf{a}}$ is the mass center accln.

$\bar{\boldsymbol{\rho}} \times m\bar{\mathbf{a}}$ = moment of magnitude of $m\bar{\mathbf{a}}$ @ *P*

→ $m\bar{a}d$

→ $$\boxed{\Sigma M_P = \bar{I}\alpha + m\bar{a}d}$$



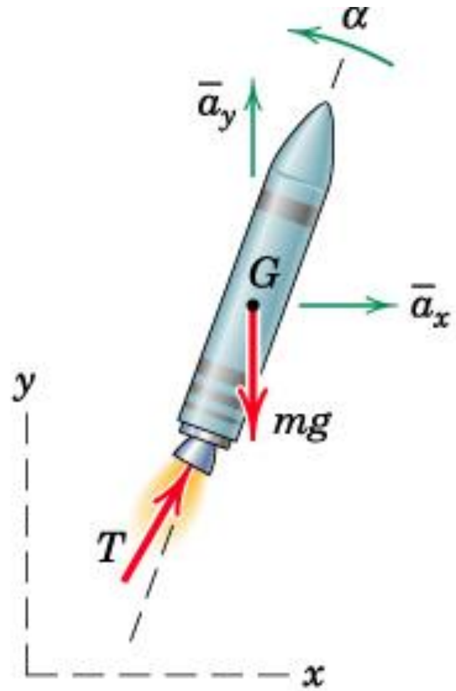
Rigid Body Kinetics :: Force/Mass/Acc

Constrained and Unconstrained Motion

:: Motion of a rigid body may be constrained or unconstrained

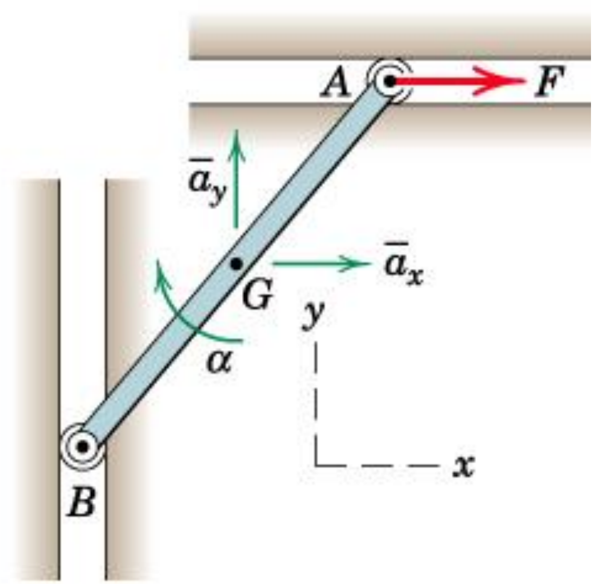
$$\Sigma \mathbf{F} = m\bar{\mathbf{a}}$$

$$\Sigma M_G = \bar{I}\alpha$$



(a) Unconstrained Motion

$\bar{a}_x$, $\bar{a}_y$, and α can be **determined independently** using plane motion eqns



(b) Constrained Motion

Kinematic relationship betn the accln comp of mass center (**linear accln**) and the **angular accln** of the bar to be determined first and then apply the plane motion eqns.

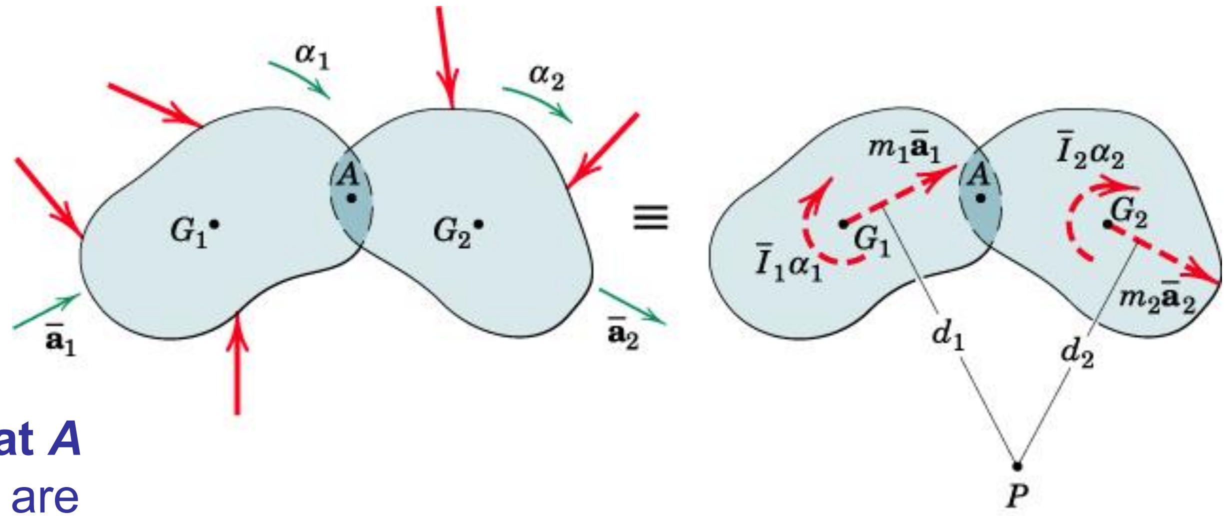
Rigid Body Kinetics :: Force/Mass/Acc

Systems of Interconnected Bodies

:: If **motion** of **connected rigid bodies** are **related kinematically**
→**analyze** the **bodies as an entire system**

$$\Sigma \mathbf{F} = \Sigma m \bar{\mathbf{a}}$$

$$\Sigma M_P = \Sigma \bar{I} \alpha + \Sigma m \bar{a} d$$



Free-Body Diagram of System ≡ Kinetic Diagram of System

Two rigid bodies hinged at *A*
Forces in the connection A are internal to the system

:: **No. of remaining unknowns** in the system **> 3** (3 eqns of plane motion insufficient)
→ Use Virtual Work method (discussed later)
→Or dismember the system and analyze each part separately

Rigid Body Kinetics :: Force/Mass/Acc

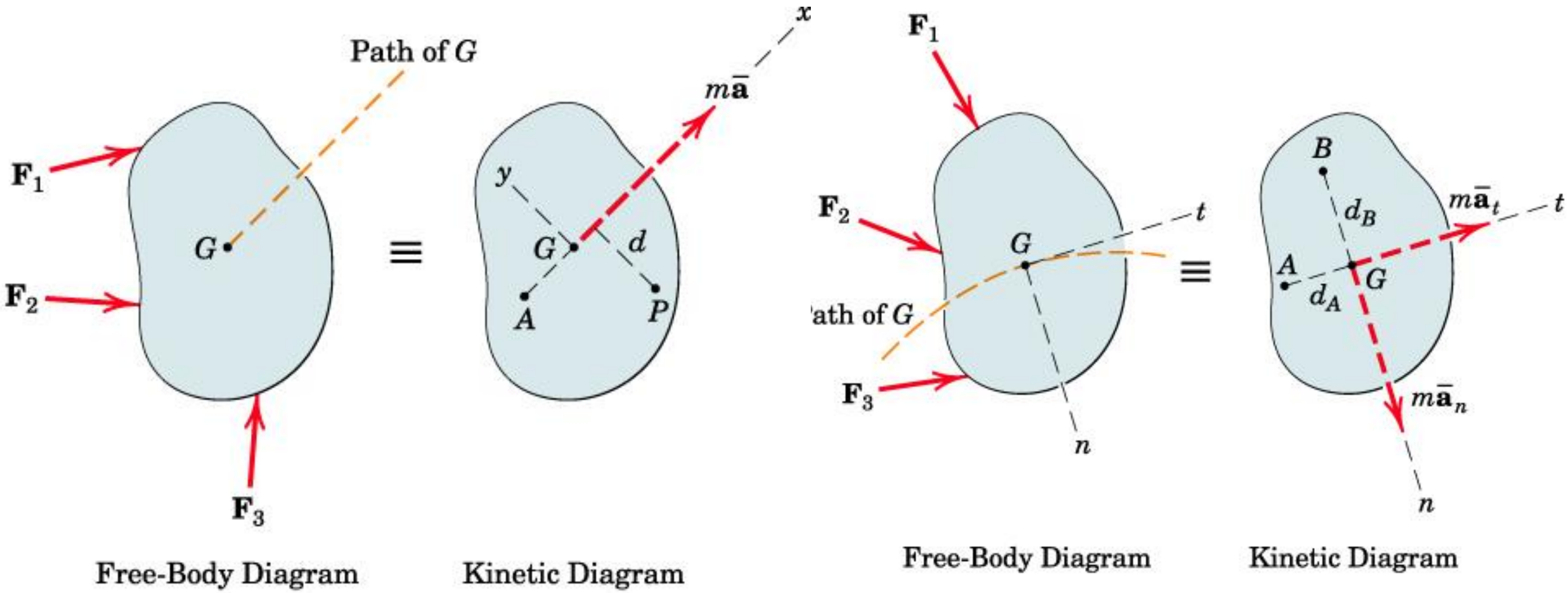
Application to three cases of rigid body motion:

Translation

→**No angular motion** of body (ω and α will be zero)
→Mass moment of inertia will not be effective

$$\Sigma \mathbf{F} = m\bar{\mathbf{a}}$$

$$\Sigma M_G = \bar{I}\alpha = 0$$



(a) Rectilinear Translation
($\alpha = 0,\ \omega = 0$)

(b) Curvilinear Translation
($\alpha = 0,\ \omega = 0$)

Example on Translation

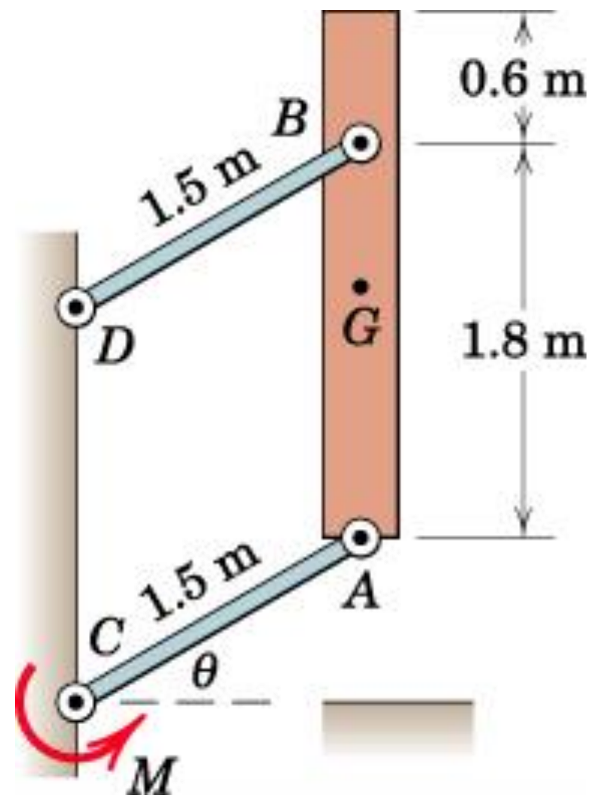
The vertical bar AB has a mass of 150 kg with center of mass G midway between the ends. The bar is elevated from rest at $\theta = 0$ by means of the parallel links of negligible mass, with a constant couple $M = 5$ kN·m applied to the lower link at C. Determine the angular acceleration α of the links as a function of θ and find the force B in the link DB at the instant when $\theta = 30°$.

Solution:

Motion of bar is curvilinear translation since the bar does not rotate.

Motion of *G (all points on AB)* is circular → **choose *n-t* coordinates**

Choosing the **reference axes coincident** with the **directions in which the comp of accln of mass center** are expressed
→ Better choice



Example on Translation

Draw the FBD and the Kinetic Diagram

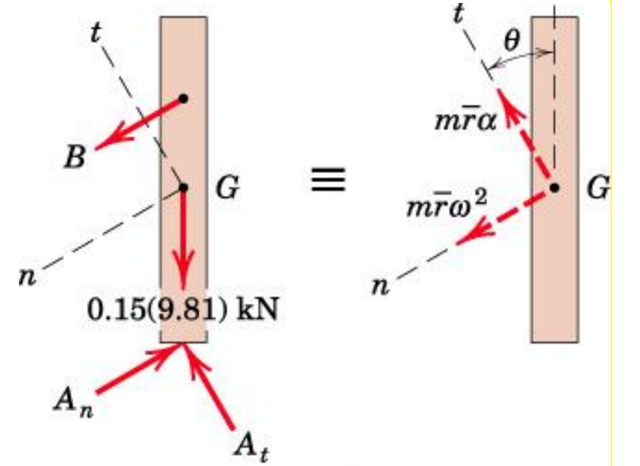
From FBD of *AC* (negligible mass→ eqn of equilibrium)

$\sum M_c = 0$ → A_t = M/1.5 = 5/1.5 = 3.33 kN

Member BD also has a negligible mass

→Two force member in equilibrium

→The force at *B* will be along the link



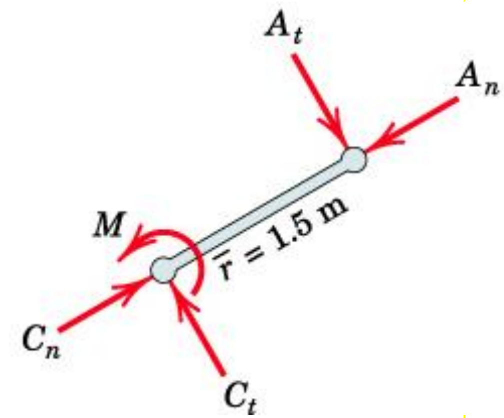
$[\Sigma F_t = m\bar{a}_t]$ $\quad 3.33 - 0.15(9.81)\cos\theta = 0.15(1.5\alpha)$

$$\alpha = 14.81 - 6.54\cos\theta \text{ rad/s}^2$$

$[\omega\, d\omega = \alpha\, d\theta]$ $\quad \int_0^\omega \omega\, d\omega = \int_0^\theta (14.81 - 6.54\cos\theta)\, d\theta$

$$\omega^2 = 29.6\theta - 13.08\sin\theta$$

θ = 30° $\quad (\omega^2)_{30^\circ} = 8.97\ (\text{rad/s})^2 \qquad \alpha_{30^\circ} = 9.15 \text{ rad/s}^2$



$$m\bar{r}\omega^2 = 0.15(1.5)(8.97) = 2.02 \text{ kN}$$

$$m\bar{r}\alpha = 0.15(1.5)(9.15) = 2.06 \text{ kN}$$

$[\Sigma M_A = m\bar{a}d]$

$$1.8\cos 30^\circ B = 2.02(1.2)\cos 30^\circ + 2.06(0.6)$$

$$B = 2.14 \text{ kN}$$

Rotation @ a Fixed Axis

Mass Moments of Inertia

- Required in rotational acceleration of any body
- **Mass *m*** of a body is a measure of **resistance to translational acceleration**
- **Area moment of inertia** is a measure of the **distribution of area @ the axis**
- Mass Moment of Inertia *I* is a measure of resistance to rotational acceleration of the body

Mass moment of inertia of the body @ *O-O*:

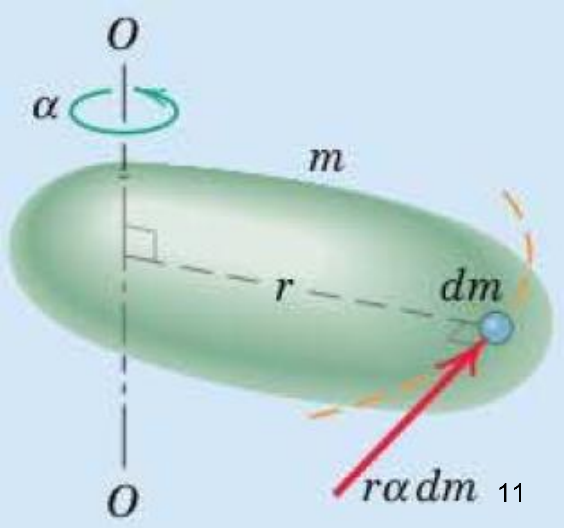
$$\boxed{I = \int r^2\, dm}$$

$$I = \Sigma r_i^2 m_i$$

$$I = \rho \int r^2\, dV$$

ρ = constant throughout the body

Units of Mass moment of inertia: kg m^2

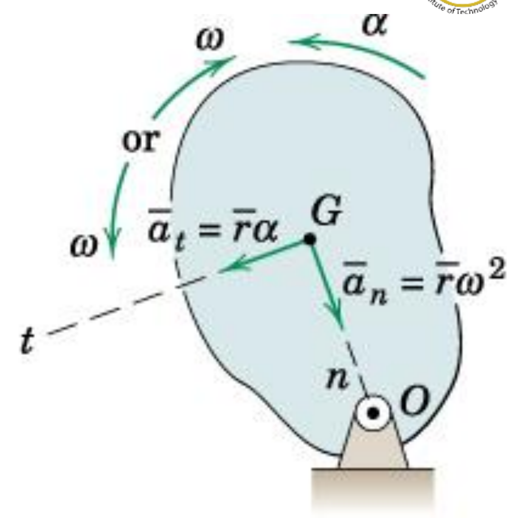


Plane Kinetics of Rigid Bodies

Fixed Axis Rotation

- All points in body move in a circular path @ rotation axis
- All lines of the body have the same ω and α

Accln components of mass center: $\bar{a}_n = \bar{r}\omega^2$ and $\bar{a}_t = \bar{r}\alpha$



Fixed-Axis Rotation

Two scalar comp of force eqns:

$$\boxed{\begin{aligned} \Sigma\mathbf{F} &= m\bar{\mathbf{a}} \\ \Sigma M_G &= \bar{I}\alpha \end{aligned}}$$

$\Sigma F_n = m\,\bar{r}\,\omega^2$ and $\Sigma F_t = m\,\bar{r}\,\alpha$

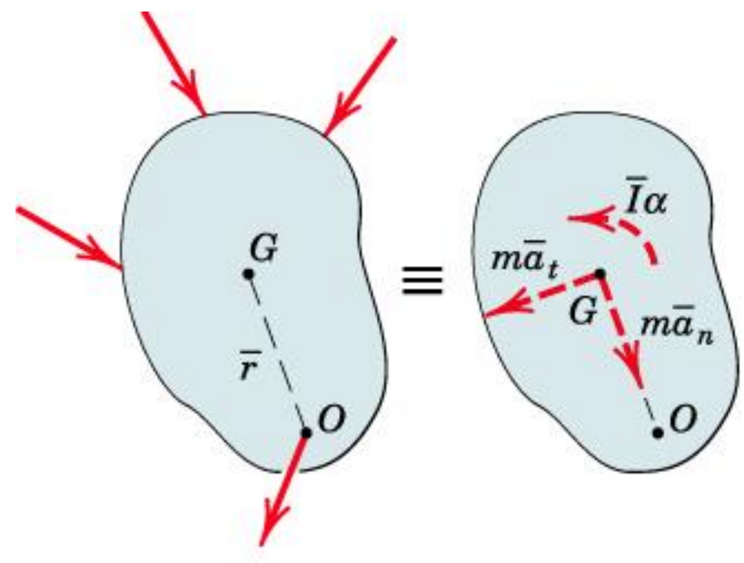
Moment of the resultants @ rotn axis O:

$$\Sigma M_O = \bar{I}\alpha + m\bar{a}_t\bar{r}$$

Using parallel axis theorem: $I_O = \bar{I} + m\bar{r}^2$

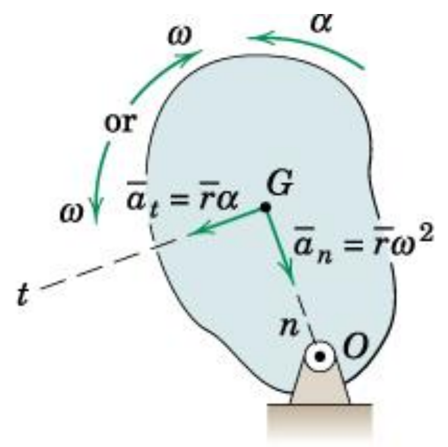
$$\rightarrow \Sigma M_O = (I_O - m\bar{r}^2)\alpha + m\bar{r}^2\alpha = I_O\alpha$$

$$\boxed{\Sigma M_O = I_O\alpha}$$



Free-Body Diagram Kinetic Diagram

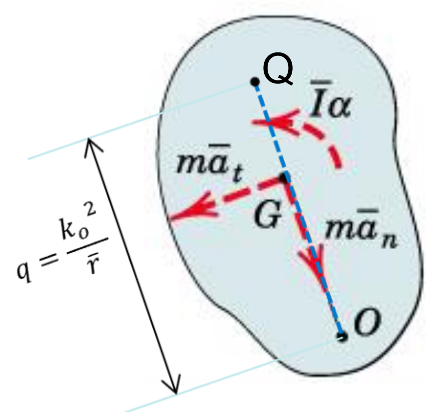
Plane Kinetics of Rigid Bodies



Fixed-Axis Rotation

Center of Percussion

- There will be one point Q along line OG, with respect to point Q resultant moment will be zero. This point is called Center of Percussion.
- Using Parallel axis theorem we have $I_o = \bar{I} + m\bar{r}^2$
- Radius of Gyration @ O: $k_o = \sqrt{I_o/m}$
- Location of Point Q: $q = \frac{{k_o}^2}{\bar{r}}$



Resultant of all the moments about Q is zero

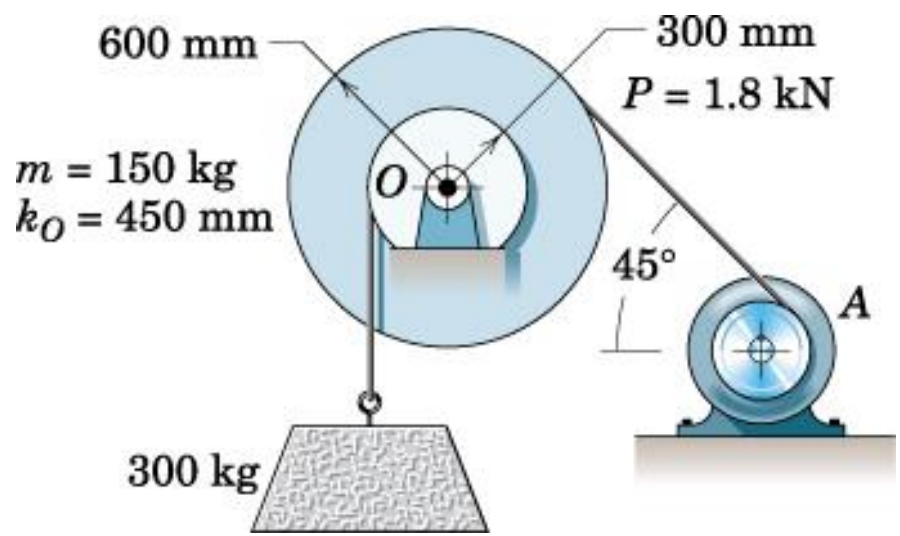
$$\begin{aligned}\sum M_Q &= \bar{I}\alpha + \overrightarrow{\boldsymbol{r_{G/Q}}} \times m\overrightarrow{\boldsymbol{a_t}} = \bar{I}\alpha - (q - \bar{r})m\bar{r}\alpha \\ &= (I_o - m\bar{r}^2)\alpha - mq\bar{r}\alpha + m\bar{r}^2\alpha \\ &= I_o\alpha - m{k_o}^2\alpha = 0\end{aligned}$$

Plane Kinetics of Rigid Bodies

Fixed Axis Rotation

Example: A concrete block is lifted by hoisting mechanism in which the cables are securely wrapped around the respective drums. The drums are fastened together and rotate as a single unit @ their mass center at *O*. Combined mass of drum is 150 kg, and radius of gyration @ *O* is 450 mm. A constant tension of 1.8 kN is maintained in the cable by the power unit at *A*. Determine the vertical accln of the block and the resultant force on the bearing at *O*.

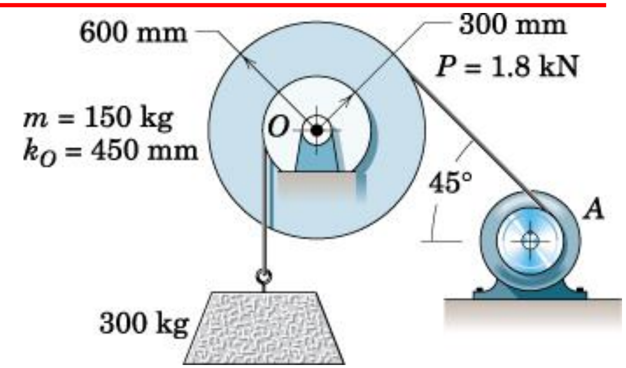
Solution:
Draw the FBD and Kinetic Diagrams



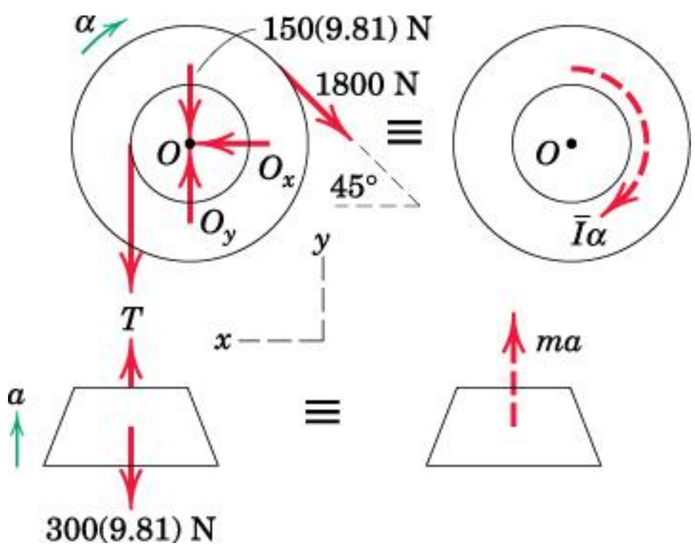
Plane Kinetics of Rigid Bodies

Force, Mass, and Acceleration

Fixed Axis Rotation

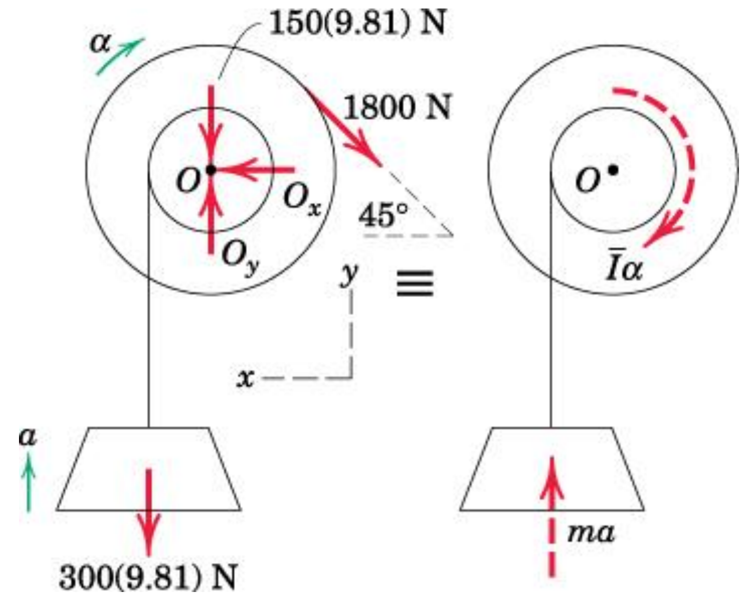


Solution: Two ways to draw the FBD and KD



KD: Resultant of the force system on the drums for centroidal rotation will be the couple $\bar{I}\alpha = I_O \alpha$

T will come into picture
→ more calculations

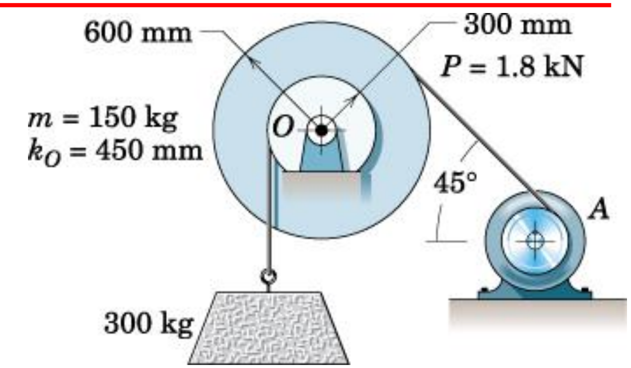


T can be eliminated by drawing FBD of the entire system.

Plane Kinetics of Rigid Bodies

Force, Mass, and Acceleration

Fixed Axis Rotation



Solution:

Resultant of the force system will be the couple $\bar{I}\alpha$ plus moment due to *ma* of the block

$I = k^2 m \rightarrow \bar{I} = I_o = (0.45)^2 150 = 30.4\ \mathrm{kgm}^2$

$[\Sigma M_O = \bar{I}\alpha + m\bar{a}d]$

$1800(0.6) – 300(9.81)(0.3) = 30.4\alpha + 300a(0.3)$

We know $a = r\,\alpha \rightarrow \alpha = a/0.3$

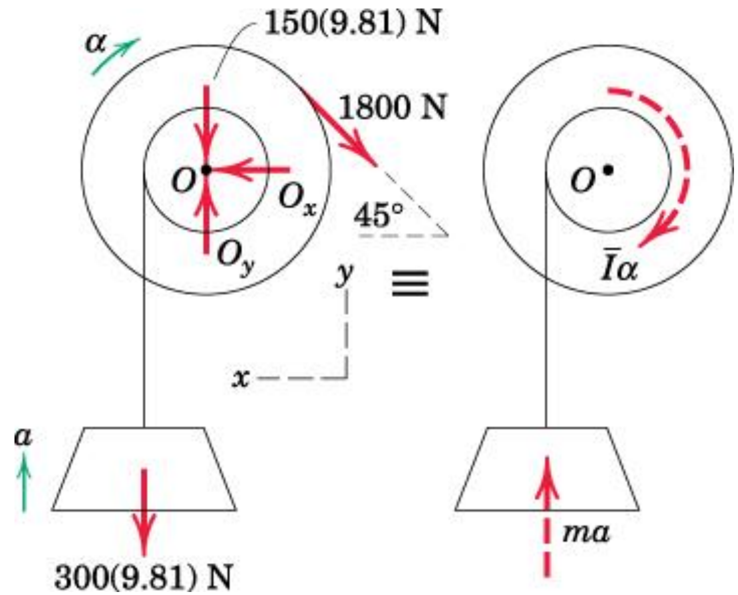
$\rightarrow$ $a = 1.031\ \mathrm{m/s^2}$ (and $\alpha = 3.44\ \mathrm{rad/s^2}$)

$[\Sigma F_y = \Sigma m\bar{a}_y]$

O_y - 150(9.81) - 300(9.81) - 1800sin45 = 150(0) + 300(1.031) $\rightarrow$ $O_y = 6000$ N

$[\Sigma F_x = \Sigma m\bar{a}_x]$

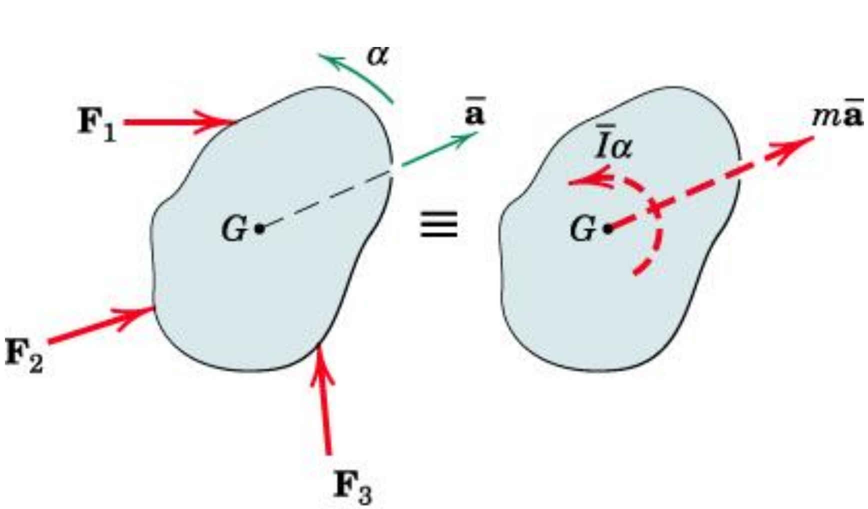
O_x – 1800cos45 = 0 $\rightarrow$ $O_x = 1273$ N



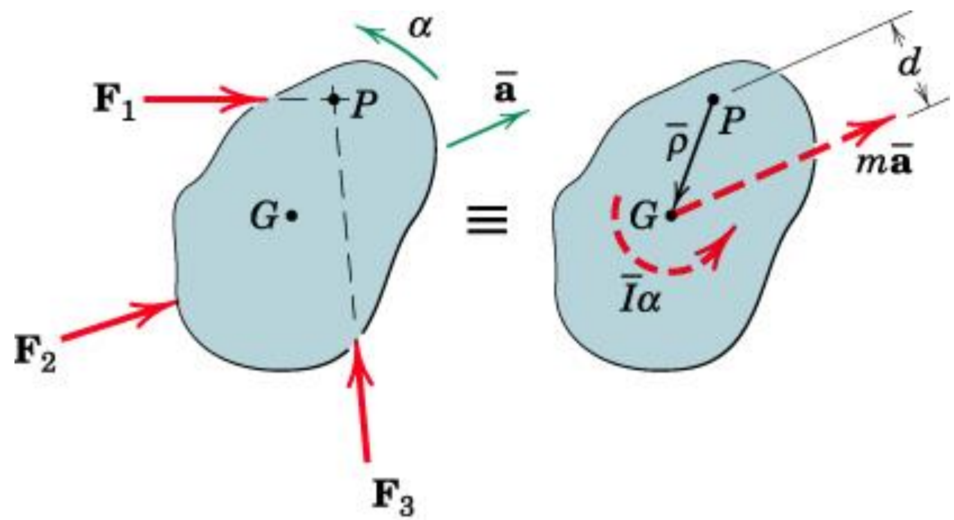
Rigid Body Kinetics :: Force/Mass/Acc

General Plane Motion

•Combines translation and rotation



Free-Body Diagram Kinetic Diagram



Free-Body Diagram Kinetic Diagram

$$\Sigma \mathbf{F} = m\bar{\mathbf{a}}$$
$$\Sigma M_G = \bar{I}\alpha$$

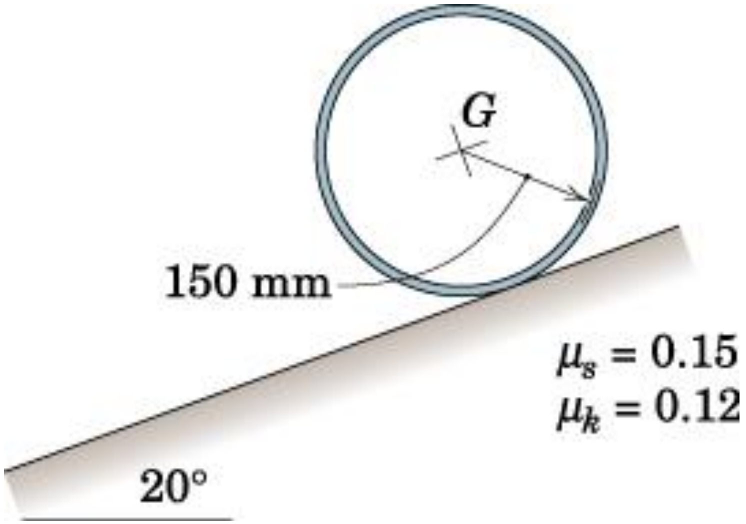
$$\Sigma M_P = \bar{I}\alpha + m\bar{a}d$$

Choice of moment eqn. :: (a) @ mass centre G

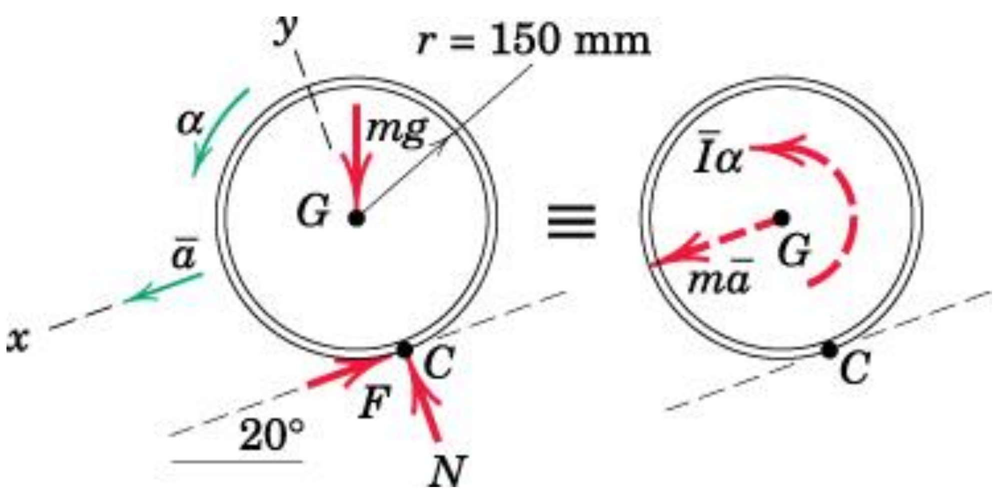
:: (b) @ point P with known acc

Example (1) on general motion

A metal hoop with radius r = 150 mm is released from rest on the 20^o incline. Coefficients of static and kinetic friction are given. Determine the angular accleration α of the hoop and the time t for the hoop to move a distance of 3 m down the incline.



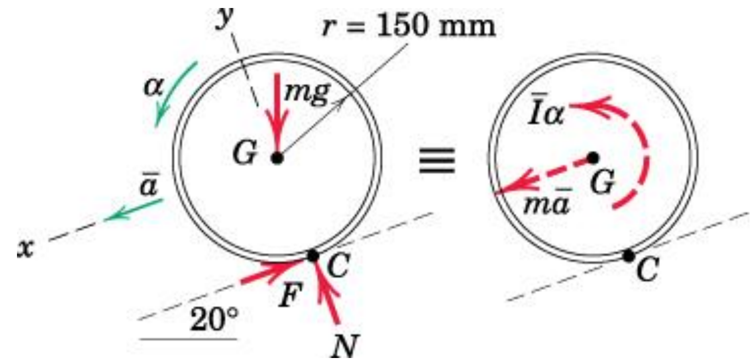
Solution: Draw the FBD and the KD



Example (1) on general motion

If the wheel slips when it rolls, $a \neq r\alpha$

Check whether the hoop slips or not when it rolls



Assume that the hoop rolls without slipping

→ $\bar{a} = r\,\alpha$ (Pure rolling)

$$\Sigma \mathbf{F} = m\bar{\mathbf{a}}$$
$$\Sigma M_G = \bar{I}\alpha$$

$$[\Sigma F_x = m\bar{a}_x] \qquad mg \sin 20° - F = m\bar{a}$$
$$[\Sigma F_y = m\bar{a}_y = 0] \qquad N - mg \cos 20° = 0$$
$$[\Sigma M_G = \bar{I}\alpha] \qquad Fr = mr^2\alpha$$

Eliminating F and substituting $\bar{a} = r\alpha$ → $\bar{a} = \frac{g}{2} \sin 20°$ → $\bar{a} = 1.678$ m/s^2

→ $\bar{a}$ is independent of both m and r

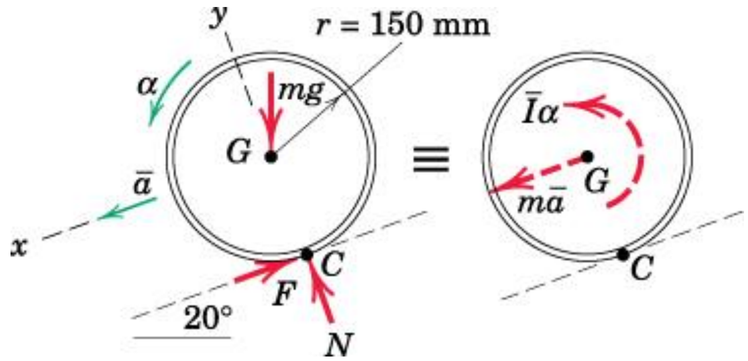
Alternatively we may use: $[\Sigma M_C = \bar{I}\alpha + m\bar{a}d]$

$mgr \sin 20° = mr^2 \frac{\bar{a}}{r} + m\bar{a}r$ → same relation → $\bar{a} = \frac{g}{2} \sin 20°$

Check for the assumption of no slipping:

Calculate F and N, and compare F with its limiting value

Example (1) on general motion



Check for the assumption of no slipping

$mg \sin 20° - F = m\bar{a}$ $N - mg \cos 20° = 0$

→ $F = mg \sin 20° - m\dfrac{g}{2} \sin 20° = 0.1710mg$

$N = mg \cos 20° = 0.940mg$

Maximum possible friction force is:

$[F_{\max} = \mu_s N]$ $F_{\max} = 0.15(0.940mg) = 0.1410mg$

Since the limiting value $F_{max} < F$

→ The assumption of pure rolling was incorrect. The hoop slips as it rolls → $\bar{a} \neq r\alpha$

→ The friction force becomes the kinetic value

$[F = \mu_k N]$ $F = 0.12(0.940mg) = 0.1128mg$ Repeat the calculations with this F

$[\Sigma F_x = m\bar{a}_x]$ $mg \sin 20° - 0.1128mg = m\bar{a}$ → $\bar{a} = 2.25$ m/s^2

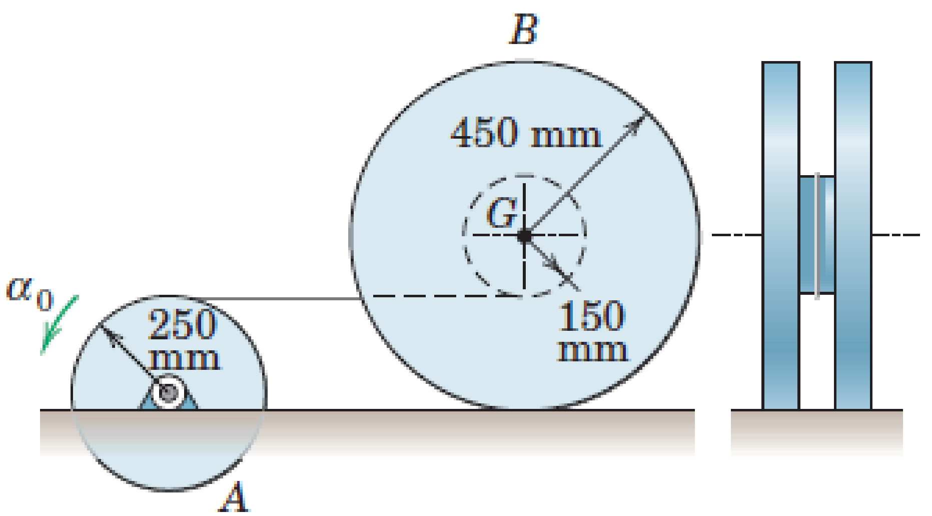
$[\Sigma M_G = \bar{I}\alpha]$ $0.1128mg(r) = mr^2\alpha$

→ $\alpha = 7.37$ rad/s^2 (α dependent on r but not on m)

$[x = \frac{1}{2}at^2]$ $t = \sqrt{\dfrac{2x}{\bar{a}}}$ at $x = 3$ m → $t = 1.633$ s

Example (2) on general motion

The drum A is given a constant angular acceleration α_0 of 3 rad/s^2 and causes the 70-kg spool B to roll on the horizontal surface by means of the connecting cable, which wraps around the inner hub of the spool. The radius of gyration $\bar{k}$ of the spool about its mass center G is 250 mm, and the coefficient of static friction between the spool and the horizontal surface is 0.25. Determine the tension T in the cable and the friction force F exerted by the horizontal surface on the spool.



Example (2) on general motion

- Solution :: FBD and KD

Horizontal component of acceleration of point D

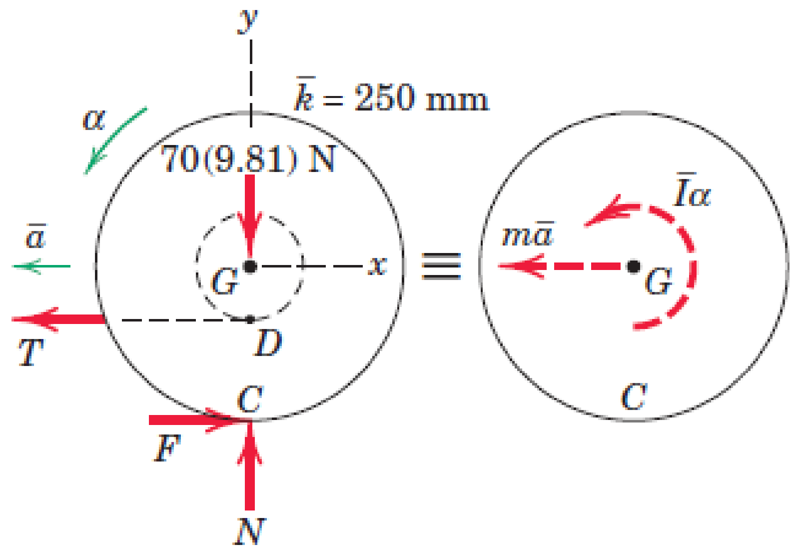
$$a_t = r\alpha = 0.25(3) = 0.75 \text{ m/s}^2$$

Assuming rolling without slipping,

$$\alpha = (a_D)_x/\overline{DC} = 0.75/0.30 = 2.5 \text{ rad/s}^2$$

Acceleration of mass centre G,

$$\bar{a} = r\alpha = 0.45(2.5) = 1.125 \text{ m/s}^2$$



The correct direction of the friction force may be assigned in this problem by observing from both diagrams that with counterclockwise angular acceleration, a moment sum about point *G (and also about point D) must be* counterclockwise

$[\Sigma F_x = m\bar{a}_x]$ $\qquad F - T = 70(-1.125)$

$[\Sigma F_y = m\bar{a}_y]$ $\qquad N - 70(9.81) = 0 \qquad N = 687$ N

$[\Sigma M_G = \bar{I}\alpha]$ $\qquad F(0.450) - T(0.150) = 70(0.250)^2(2.5)$

Solving (a) and (b) simultaneously gives

$$F = 75.8 \text{ N} \qquad \text{and} \qquad T = 154.6 \text{ N}$$

Example (2) on general motion

To establish the validity of our assumption of no slipping, we see that the surfaces are capable of supporting a maximum friction force $F_{\max} = \mu_s N = 0.25(687) = 171.7$ N. Since only 75.8 N of friction force is required, we conclude that our assumption of rolling without slipping is valid.

Plane Kinetics of Rigid Bodies

Work and Energy

Advantages of Work Energy Method

•These principles are especially useful in describing motion resulting from the cumulative effect of forces acting through distances.

•If the forces are conservative: velocity changes can be determined by analyzing the energy conditions only at the beginning and at the end of the motion interval.

•For finite displacements: no need to compute acceleration; leads directly to velocity changes as functions of forces, which do work.

•Involves only those forces, which do work, and thus, produces change in magnitudes of velocities.

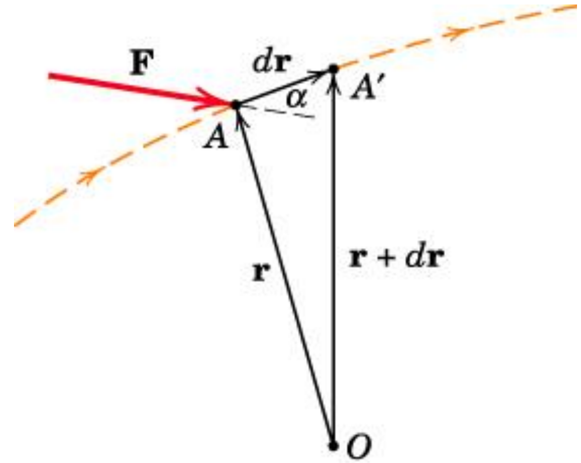
→ Simplifies calculations

Work and Energy

Work done by force **F**:

$$U = \int \mathbf{F} \cdot d\mathbf{r}$$ or $$U = \int (F \cos \alpha)\, ds$$

ds is the magnitude of the vector displacement *d***r**



Work done by couple *M*:

Work done by both forces (*F*) during translation part of motion cancel each other (opposite dirn)

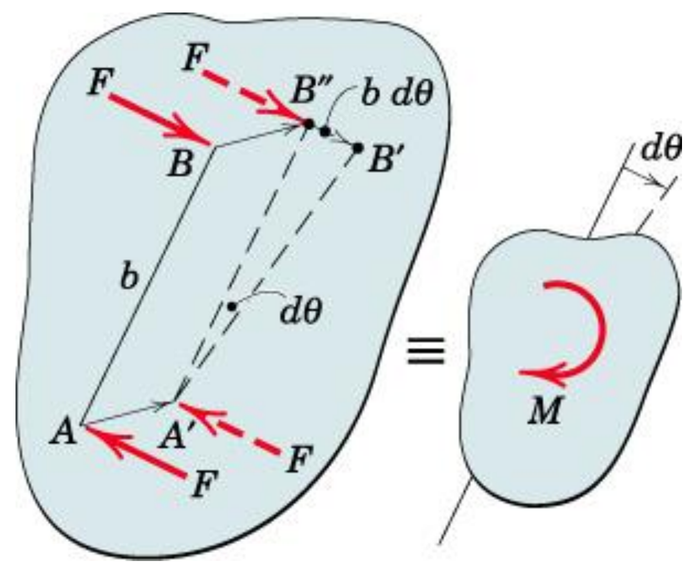
∴ Net work done will be the work done by the couple:

$$dU = F(b\, d\theta) = M\, d\theta$$

(due to rotational part of motion)

During finite rotation, work done:

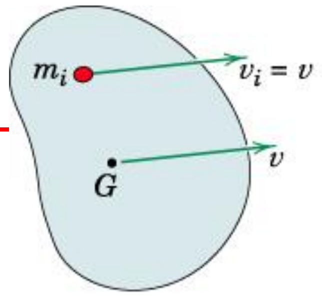
$$U = \int M\, d\theta$$



Plane Kinetics of Rigid Bodies

Work and Energy

Kinetic Energy: Three classes of rigid body plane motion



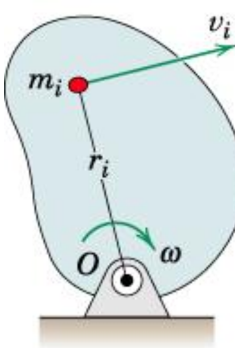
(a) Translation

Translation

•All particles will have same velocity

For entire body: $T = \sum ½\, m_i v^2$ → (both rectilinear and curvilinear)

$$\boxed{T = \frac{1}{2} m v^2}$$



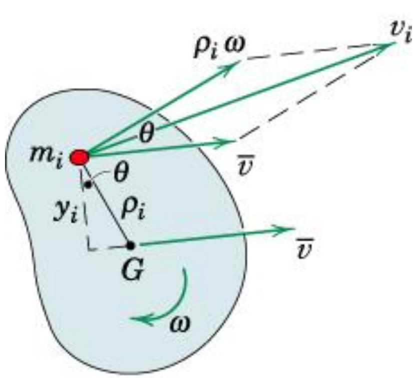
(b) Fixed-Axis Rotation

Fixed Axis Rotation

For a particle m_i : $T_i = ½\, m_i (r_i \omega)^2$

For entire body: $T = ½\, \omega^2 \sum m_i r_i^2 = ½\, \omega^2 I_o$

$$\boxed{T = \frac{1}{2} I_O \omega^2}$$



(c) General Plane Motion

General Plane Motion

$$T = \Sigma \frac{1}{2} m_i v_i^2 = \Sigma \frac{1}{2} m_i (\bar{v}^2 + \rho_i^2 \omega^2 + 2\bar{v} \rho_i \omega \cos\theta)$$

Third summation: $\omega \bar{v}\, \Sigma m_i \rho_i \cos\theta = \omega \bar{v}\, \Sigma m_i y_i = 0$

→ $T =$ $\frac{1}{2} \bar{v}^2 \Sigma m_i + \frac{1}{2} \omega^2 \Sigma m_i \rho_i^2$

$$\boxed{T = \frac{1}{2} m \bar{v}^2 + \frac{1}{2} \bar{I} \omega^2}$$

Also, KE in terms of rotational vel @ the instantaneous center C of zero vel

$$\boxed{T = \frac{1}{2} I_C \omega^2}$$

Plane Kinetics of Rigid Bodies

Work and Energy

Potential Energy and Work-Energy Equation

Work-Energy relation for the motion of a general system of particles:

$$\boxed{T_1 + U_{1\text{-}2} = T_2}$$

($U_{1\text{-}2}$ is work done by all external forces)

$$\boxed{T_1 + V_1 + U'_{1\text{-}2} = T_2 + V_2}$$

($U'_{1\text{-}2}$ is work done by all external forces other than weight and spring forces, which are taken care of by means of potential energy rather than work)

•In case of interconnected system of rigid bodies, the work-energy equations include the effect of stored elastic energy in the connections.

•$U'_{1\text{-}2}$ includes the negative work of internal friction forces.

•Work-energy method is most useful for analyzing conservative systems of interconnected bodies, where energy loss due to -ve work of friction is negligible.

Plane Kinetics of Rigid Bodies

Work and Energy

Power

Power can also be expressed as rate of change of total mechanical energy of the system of rigid bodies:

Work-energy relation for an infinitesimal displacement:

$$\boxed{T_1 + V_1 + U'_{1\text{-}2} = T_2 + V_2}$$ → $dU' = dT + dV$

dU' is the work done by the active forces and couples applied to the bodies

Dividing by *dt*: $$P = \frac{dU'}{dt} = \dot{T} + \dot{V} = \frac{d}{dt}(T + V)$$

→ Total power developed by the active forces and couples equals the rate of change of the total mechanical energy of the bodies or system of bodies

Since

$$\boxed{T = \frac{1}{2}m\bar{v}^2 + \frac{1}{2}\bar{I}\omega^2}$$ →

$$\dot{T} = \frac{dT}{dt} = \frac{d}{dt}\left(\frac{1}{2}m\bar{\mathbf{v}}\cdot\bar{\mathbf{v}} + \frac{1}{2}\bar{I}\omega^2\right) = \frac{1}{2}m(\bar{\mathbf{a}}\cdot\bar{\mathbf{v}} + \bar{\mathbf{v}}\cdot\bar{\mathbf{a}}) + \bar{I}\omega\dot{\omega}$$

$$= m\bar{\mathbf{a}}\cdot\bar{\mathbf{v}} + \bar{I}\alpha(\omega) = \mathbf{R}\cdot\bar{\mathbf{v}} + \bar{M}\omega$$

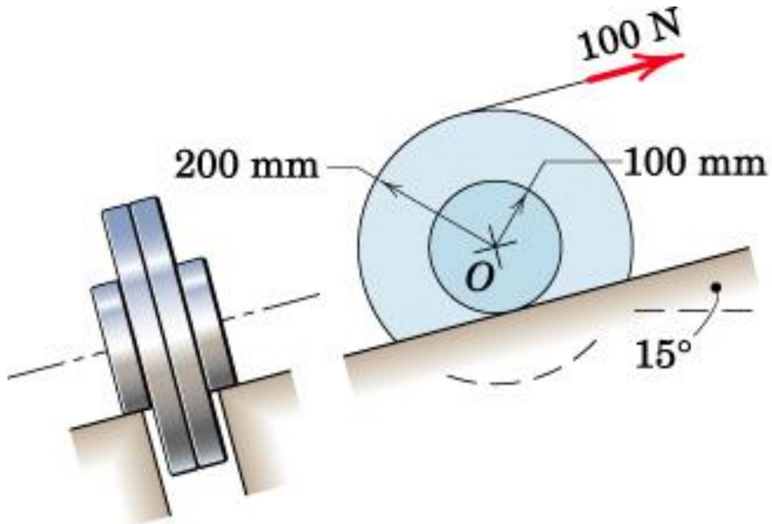
R is resultant of all forces acting on body, & $\bar{M}$ is the resultant moment @ *G* of all forces

Dot product accounts for the case of curvilinear motion of the mass center, where $\bar{\mathbf{v}}$ and $\bar{\mathbf{a}}$ are not in the same direction.

Work and Energy

Example

The wheel rolls up the incline on its hubs without slipping and is pulled by the 100-N force applied to the cord wrapped around its outer rim. If the wheel starts from rest, compute its angular velocity ω after its center has moved a distance of 3 m up the incline. The wheel has a mass of 40 kg with center of mass at O and has a centroidal radius of gyration of 150 mm. Determine the power input from the 100-N force at the end of the 3-m motion interval.

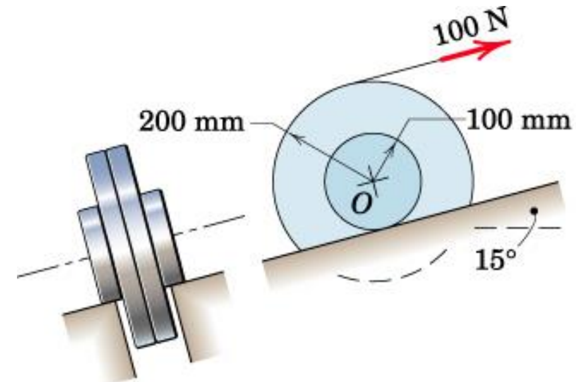


Solution

- Wheel has a general plane motion
- Draw the FBD of the wheel

Plane Kinetics of Rigid Bodies

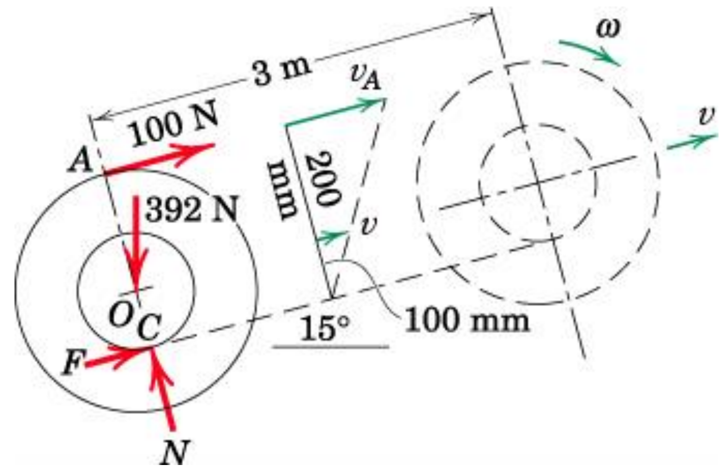
Solution



- Draw the FBD of the wheel
- Only 100 N and 40x9.81 = 392 N forces do work

C is the instantaneous center of zero velocity

→ Vel of point *A*: v_A = [(200+100)/100]v = 3v
→Point *A* on the cord moves a dist of:
(200+100)/100 = 3 times that of *O* = 3x3 = 9m



Including the effect of weight in *U* term:

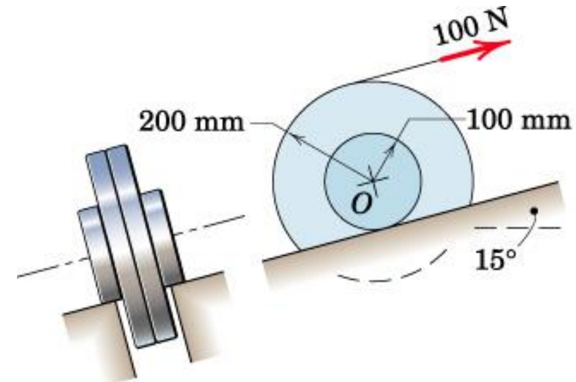
The work done by the wheel:
$U_{1\text{-}2}$ = 100(9) – 392sin15(3) = 595 J

For general plane motion, KE: $\left[T = \frac{1}{2}m\bar{v}^2 + \frac{1}{2}\bar{I}\omega^2\right]$

T_1 = 0; T_2 = ½ (40)(0.1ω)^2 + ½ (40)(0.15)$^2\omega^2$ = 0.65 ω^2

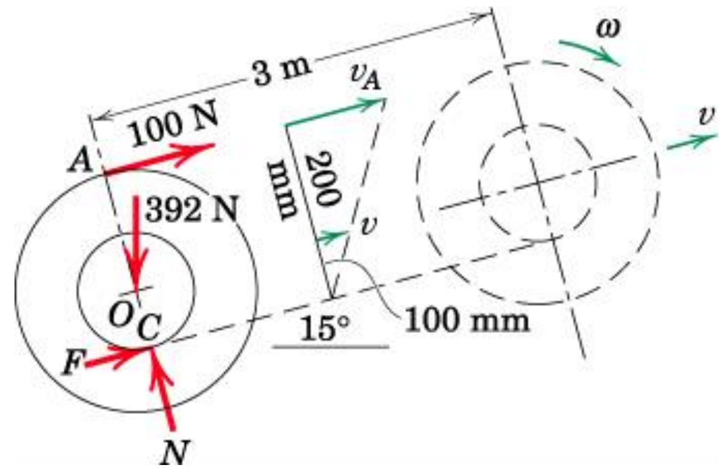
since vel of the center of the wheel $\bar{v}$ = $r\,\omega$ = 0.1 ω

Work and Energy



Solution

Work-energy eqn: $[T_1 + U_{1\text{-}2} = T_2]$
$0 + 595 = 0.65\,\omega^2$
→ $\omega = 30.3$ rad/s



We may also calculate the KE of the wheel using KE in terms of rotational vel @ the instantaneous center C of zero vel

$[T = \frac{1}{2} I_C \omega^2]$ $\quad I_c = \bar{I} + m|OC|^2$ and $\bar{I} = I_o = mk_o^2$

$$T = \frac{1}{2} 40[(0.15)^2 + (0.10)^2]\omega^2 = 0.650\omega^2$$

→ Same relation.

Power input from 100 N force when $\omega = 30.3$ rad/s:
$P = \mathbf{F} \cdot \mathbf{v}$
→ $P = 100(0.3)(30.3) = 908$ W

Plane Kinetics of Rigid Bodies

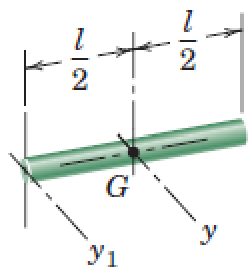
Work and Energy

Example

In the mechanism shown, each of the two wheels has a mass of 30 kg and a centroidal radius of gyration of 100 mm. Each link *OB* has a mass of 10 kg and may be treated as a slender bar. The 7-kg collar at *B* slides on the fixed vertical shaft with negligible friction. The spring has a stiffness $k = 30$ kN/m and is contacted by the bottom of the collar when the links reach the horizontal position. If the collar is released from rest at the position $\theta = 45°$ and if friction is sufficient to prevent the wheels from slipping, determine (*a*) the velocity v_B of the collar as it first strikes the spring and (*b*) the maximum deformation *x* of the spring.

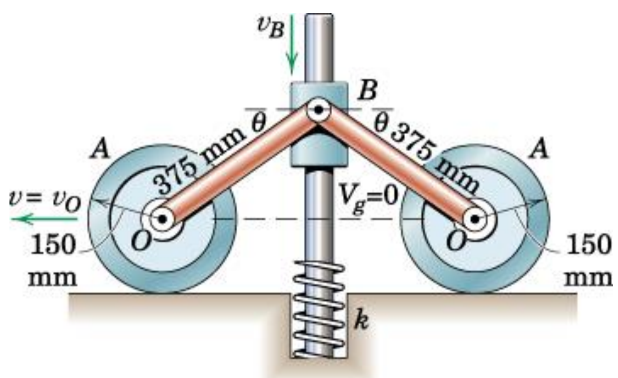
Solution

- Plane Motion
- Conservative since friction forces can be neglected
- Choosing the datum for zero gravitational potential energy v_g through *O*.



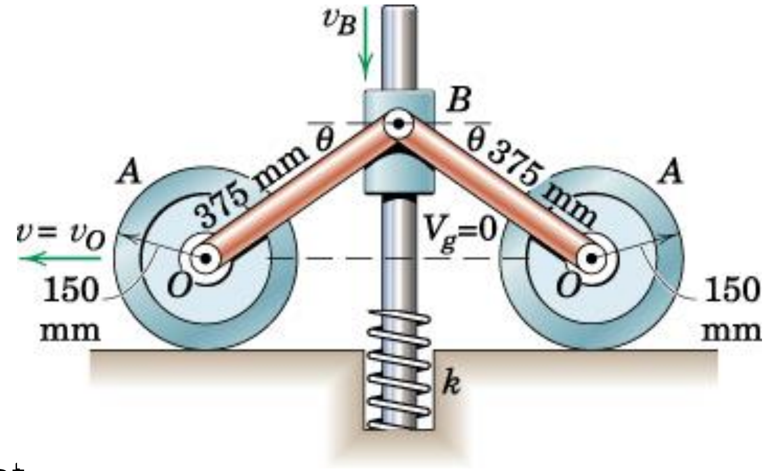
$$I_{yy} = \frac{1}{12}ml^2$$

$$I_{y_1y_1} = \frac{1}{3}ml^2$$



Plane Kinetics of Rigid Bodies

Work and Energy



Solution

Let us define three states:
at θ = 45°, θ = 0°, & max spring deflection

At θ = 45° → the wheel starts from rest
At θ = 0° → the wheel momentarily comes to rest
→ For the interval from θ = 45° to θ = 0°, the initial and final KE of the wheels = 0
At State 2: Each link is rotating @ O → Total KE:

$$T_2 = [2(\tfrac{1}{2} I_O \omega^2)]_{\text{links}} + [\tfrac{1}{2} m v^2]_{\text{collar}} = \frac{1}{3} 10(0.375)^2 \left(\frac{v_B}{0.375}\right)^2 + \frac{1}{2} 7 v_B{}^2 = 6.83 v_B{}^2$$

During this interval, the collar B drops a distance 0.375/√2 = 0.265 m

→ PE: $$V_1 = 2(10)(9.81)\frac{0.265}{2} + 7(9.81)(0.265) = 44.2 \text{ J} \qquad V_2 = 0$$

V_2 is zero since the collar and the links reaches the datum.
There are no active forces that are doing work (other than the weights, which are included in PE) → $U'_{1\text{-}2}$ = 0 → Work-Energy equation →

$$[T_1 + V_1 + U'_{1\text{-}2} = T_2 + V_2] \qquad 0 + 44.2 + 0 = 6.83 v_B{}^2 + 0 \qquad v_B = 2.54 \text{ m/s}$$

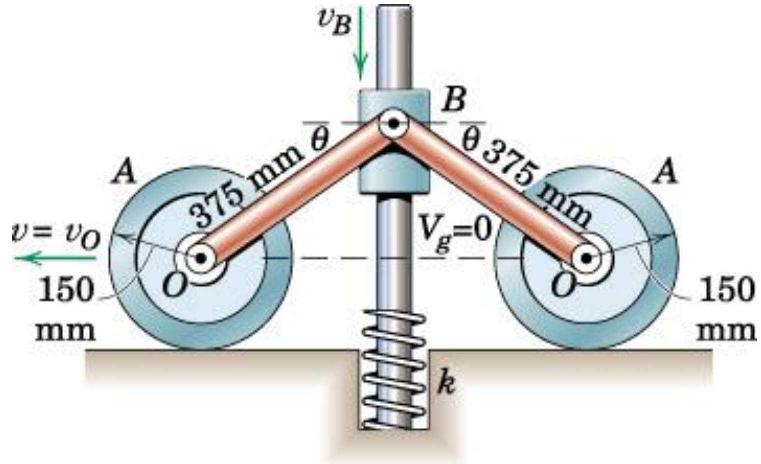
Plane Kinetics of Rigid Bodies

Work and Energy

Example

Solution

At the third state (max deflection of spring),
all parts of the system are momentarily at rest.
→ KE, $T_3 = 0$



Using the Work-Energy equation between states 1 and 3

$$[T_1 + V_1 + U'_{1\text{-}3} = T_3 + V_3]$$

$$0 + 2(10)(9.81)\,\frac{0.265}{2} + 7(9.81)(0.265) + 0$$
$$= 0 - 2(10)(9.81)\left(\frac{x}{2}\right) - 7(9.81)x + \frac{1}{2}(30)(10^3)x^2$$

→ Maximum deformation of the spring:
  $x = 60.1$ mm (positive root)

Try solving this problem without using the work-energy equation.

Rigid Body Kinetics :: Virtual Work

Work-energy relation for an infinitesimal displacement:

dU’ = dT + dV (dU’ :: total work done by all active forces)

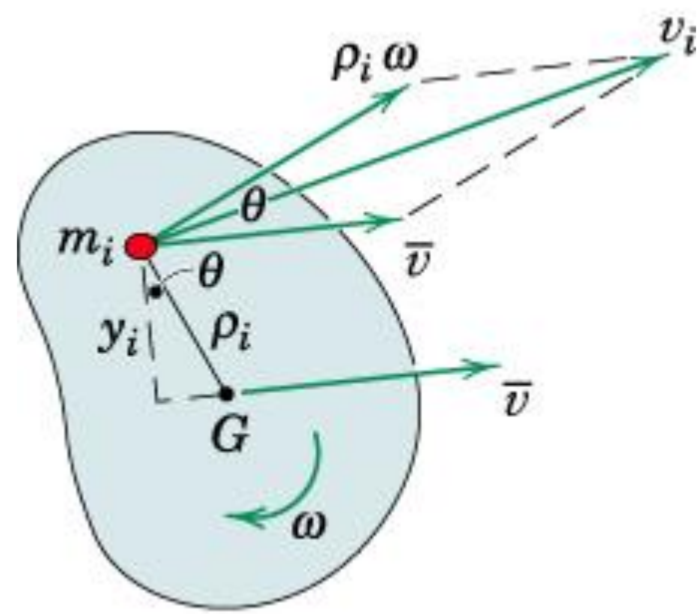
For interconnected systems, differential change in KE for the entire system:

$$dT = d(\Sigma \tfrac{1}{2} m_i \bar{v}_i^{\,2} + \Sigma \tfrac{1}{2} \bar{I}_i \omega_i^{\,2}) = \Sigma m_i \bar{v}_i \, d\bar{v}_i + \Sigma \bar{I}_i \omega_i \, d\omega_i$$

For each body: $m_i \bar{v}_i \, d\bar{v}_i = m_i \bar{\mathbf{a}}_i \cdot d\bar{\mathbf{s}}_i$ and $\bar{I}_i \omega_i \, d\omega_i = \bar{I}_i \alpha_i \, d\theta_i$

$d\bar{\mathbf{s}}_i$ :: infinitesimal linear disp of the center of mass

$d\theta_i$ :: infinitesimal angular disp of the body in the plane of motion

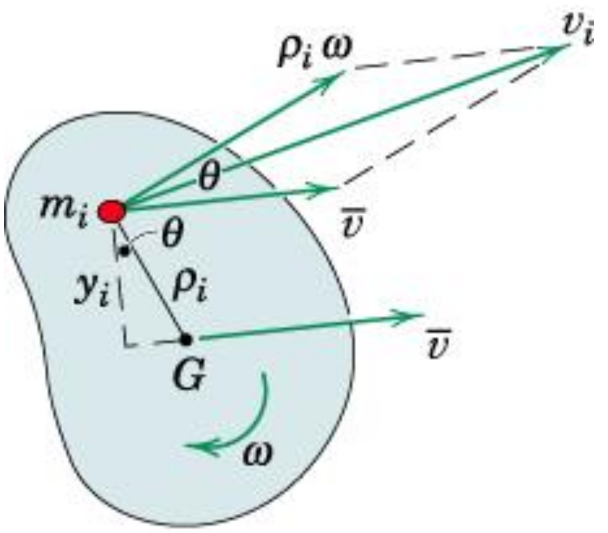


Rigid Body Kinetics :: Virtual Work

Now, $\bar{\mathbf{a}}_i \cdot d\bar{\mathbf{s}}_i$ is identical to $(\bar{a}_i)_t\, d\bar{s}_i$

α_i :: angular accln of the body

$(\bar{a}_i)_t$ :: component of $\bar{\mathbf{a}}_i$ along tangent to the curve described by mass center of the body



→ $dT = \Sigma m_i \bar{\mathbf{a}}_i \cdot d\bar{\mathbf{s}}_i + \Sigma \bar{I}_i \alpha_i \, d\theta_i$

→ $dT = \Sigma \mathbf{R}_i \cdot d\bar{\mathbf{s}}_i + \Sigma \mathbf{M}_{G_i} \cdot d\boldsymbol{\theta}_i$

$\mathbf{R}_i$ :: resultant force acting on i^{th} body

$\mathbf{M}_{Gi}$ :: resultant couple acting on i^{th} body

$d\boldsymbol{\theta}_i$ :: $d\theta_i\, \mathbf{k}$

∴ **Differential change** in **kinetic energy**

= **Differential work** done by the **resultant forces** and **couples**

Rigid Body Kinetics :: Virtual Work

$dU' = dT + dV$ (dU' :: total work done by all active forces)

dV :: differential change in total V_g and V_e

$$dV = d(\Sigma m_i g h_i + \Sigma \tfrac{1}{2} k_j x_j^2) = \Sigma m_i g\, dh_i + \Sigma k_j x_j\, dx_j$$

h_i => vertical distance of the center of mass m_i above a convenient datum plane

x_i => deformation of elastic member (spring of stiffness k_j) of system

$$dT = \Sigma m_i \bar{\mathbf{a}}_i \cdot d\bar{\mathbf{s}}_i + \Sigma \bar{I}_i \alpha_i\, d\theta_i$$ (+ve for same dirn. of accn and disp)

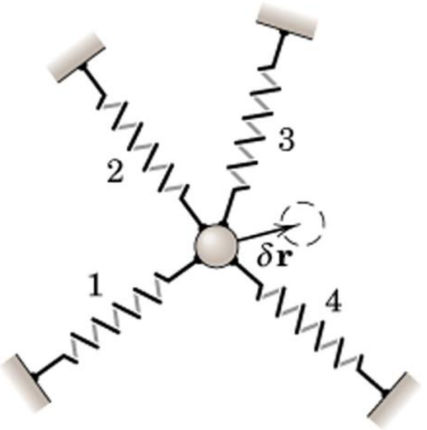
→ $$\boxed{dU' = \Sigma m_i \bar{\mathbf{a}}_i \cdot d\bar{\mathbf{s}}_i + \Sigma \bar{I}_i \alpha_i\, d\theta_i + \Sigma m_i g\, dh_i + \Sigma k_j x_j\, dx_j}$$

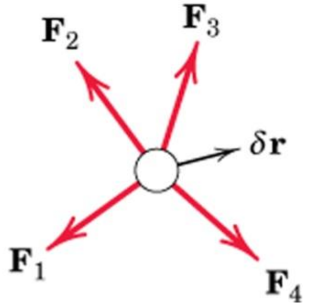
→ Direct relation between the accelerations and the active forces

Rigid Body Kinetics :: Virtual Work

- **Statics**
  - Virtual work eqn

$$\delta U = \Sigma \mathbf{F} \cdot \delta \mathbf{r} = (\mathbf{i}\, \Sigma F_x + \mathbf{j}\, \Sigma F_y + \mathbf{k}\, \Sigma F_z) \cdot (\mathbf{i}\, \delta x + \mathbf{j}\, \delta y + \mathbf{k}\, \delta z)$$
$$= \Sigma F_x\, \delta x + \Sigma F_y\, \delta y + \Sigma F_z\, \delta z = 0$$





- **Kinetics**

$$\delta U' = \Sigma m_i \bar{\mathbf{a}}_i \cdot \delta \bar{\mathbf{s}}_i + \Sigma \bar{I}_i \alpha_i\, \delta\theta_i + \Sigma m_i g\, \delta h_i + \Sigma k_j x_j\, \delta x_j$$

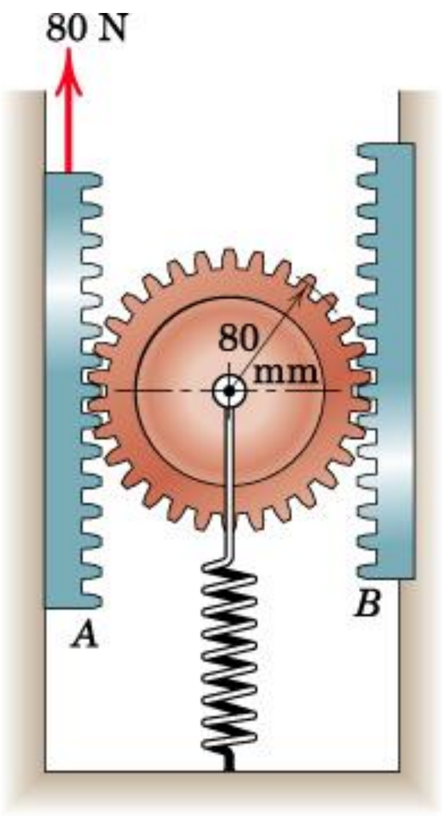
- If a rigid body is in equilibrium

total virtual work of external forces acting on the body is zero for any virtual displacement of the body

Virtual displacements should be consistent with the kinematic constraints.

Example (1) on virtual work

The movable rack A has a mass of 3 kg, and rack B is fixed. The gear has a mass of 2 kg and a radius of gyration of 60 mm. In the position shown, the spring, which has a stiffness of 1.2 kN/m, is stretched a distance of 40 mm. For the instant represented, determine the acceleration a of rack A under the action of the 80-N force. The plane of the figure is vertical.



Solution:

: Work done by the weight and spring will be accounted for in the PE terms

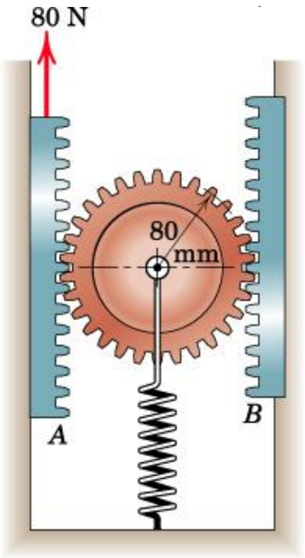
: No other external force except 80 N do any work

For an infinitesimal upward disp dx of rack A,
work done on the system: $dU' = 80\,dx$

$\rightarrow dU' = dT + dV$

$$dU' = \Sigma m_i \bar{\mathbf{a}}_i \cdot d\bar{\mathbf{s}}_i + \Sigma \bar{I}_i \alpha_i\, d\theta_i + \Sigma m_i g\, dh_i + \Sigma k_j x_j\, dx_j$$

Example (1) on virtual work



$$dU' = \Sigma m_i \bar{\mathbf{a}}_i \cdot d\bar{\mathbf{s}}_i + \Sigma \bar{I}_i \alpha_i \, d\theta_i + \Sigma m_i g \, dh_i + \Sigma k_j x_j \, dx_j$$

$$[dT = \Sigma m_i \bar{\mathbf{a}}_i \cdot d\bar{\mathbf{s}}_i + \Sigma \bar{I}_i \alpha_i \, d\theta_i]$$

$$dT_{\text{rack}} = 3a \, dx \qquad dT_{\text{gear}} = 2 \frac{a}{2} \frac{dx}{2} + 2(0.06)^2 \frac{a/2}{0.08} \frac{dx/2}{0.08} = 0.781a \, dx$$

Since disp and accln of mass center of gear will be half that of rack *A*

$$[dV = \Sigma m_i g \, dh_i + \Sigma k_j x_j \, dx_j]$$

$$dV_{\text{rack}} = 3g \, dx = 3(9.81) \, dx = 29.4 \, dx \qquad dV_{\text{gear}} = 2g(dx/2) = g \, dx = 9.81 \, dx$$

$$dV_{\text{spring}} = k_j x_j \, dx_j = 1200(0.04) \, dx/2 = 24 \, dx$$

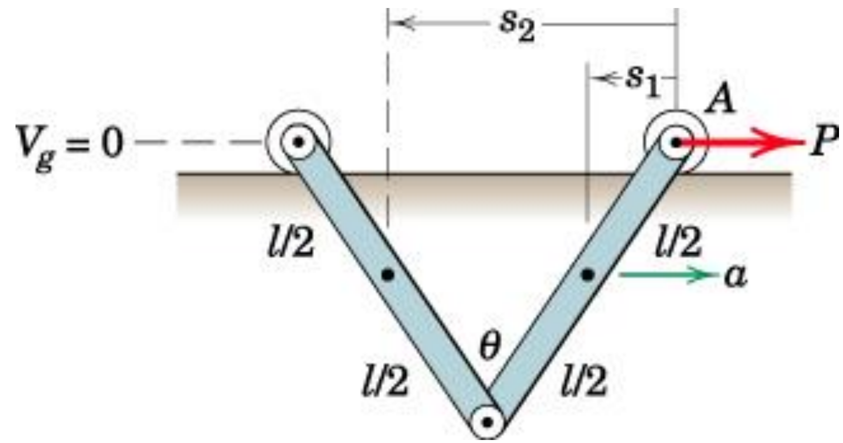
Substituting in the work-energy equation:

$$80 \, dx = 3a \, dx + 0.781a \, dx + 29.4 \, dx + 9.81 \, dx + 24 \, dx$$

→ $a = 4.43 \text{ m/s}^2$

Example (2) on virtual work

A constant force P is applied to end A of the two identical and uniform links and causes them to move to the right in their vertical plane with a horizontal acceleration a. Determine the steady-state angle θ made by the bars with one another.



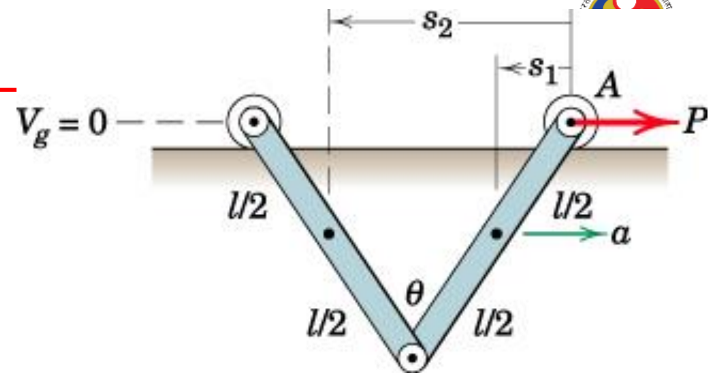
Solution:

Consider position vector s_1 and s_2 *respectively for mass centres of the bars* from A

$$\delta U' = \Sigma m_i \bar{\mathbf{a}}_i \cdot \delta \bar{\mathbf{s}}_i + \Sigma \bar{I}_i \alpha_i \, \delta\theta_i + \Sigma m_i g \, \delta h_i + \Sigma k_j x_j \, \delta x_j$$

Example (2) on virtual work

$$\delta U' = \Sigma m_i \bar{\mathbf{a}}_i \cdot \delta \bar{\mathbf{s}}_i + \Sigma \bar{I}_i \alpha_i \, \delta\theta_i + \Sigma m_i g \, \delta h_i + \Sigma k_j x_j \, \delta x_j$$



For virtual displacements in steady state,

α = 0 for both bars

$$\Sigma m \bar{\mathbf{a}} \cdot \delta \bar{\mathbf{s}} = ma(-\delta s_1) + ma(-\delta s_2) = -ma\left[\delta\left(\frac{l}{2}\sin\frac{\theta}{2}\right) + \delta\left(\frac{3l}{2}\sin\frac{\theta}{2}\right)\right] = -ma\left(l \cos\frac{\theta}{2}\,\delta\theta\right)$$

Choosing horz line through *A* as the datum for zero potential energy:

→ PE of the links: $V_g = 2mg\left(-\frac{l}{2}\cos\frac{\theta}{2}\right)$

→ Virtual change in PE: $\delta V_g = \delta\left(-2mg\frac{l}{2}\cos\frac{\theta}{2}\right) = \frac{mgl}{2}\sin\frac{\theta}{2}\,\delta\theta$

Substituting in the work-energy eqn for virtual changes:

$$0 = -mal\cos\frac{\theta}{2}\,\delta\theta + \frac{mgl}{2}\sin\frac{\theta}{2}\,\delta\theta$$

→ $\theta = 2\tan^{-1}\frac{2a}{g}$

Rigid Body Kinetics :: Impulse/Momentum

Impulse-Momentum equations are useful

- when the applied forces are expressed as functions of time
- when interaction between particles or rigid bodies occur during short periods of time, such as, impact

Linear Momentum

Linear momentum of any mass system, rigid or non-rigid: $\mathbf{G} = m\bar{\mathbf{v}}$

$\bar{\mathbf{v}}$ is the velocity if mass center

$$\Sigma\mathbf{F} = \dot{\mathbf{G}}$$

$$\mathbf{G}_1 + \int_{t_1}^{t_2} \Sigma\mathbf{F}\, dt = \mathbf{G}_2$$

F dt
= Linear Impulse

$$\Sigma F_x = \dot{G}_x \qquad \Sigma F_y = \dot{G}_y$$

$$(G_x)_1 + \int_{t_1}^{t_2} \Sigma F_x\, dt = (G_x)_2$$

$$(G_y)_1 + \int_{t_1}^{t_2} \Sigma F_y\, dt = (G_y)_2$$

Rigid Body Kinetics :: Impulse/Momentum

Angular Momentum

$\mathbf{H}_G = \Sigma \boldsymbol{\rho}_i \times m_i \mathbf{v}_i$ $\quad$ $\mathbf{H}_G = \Sigma \boldsymbol{\rho}_i \times m_i \dot{\boldsymbol{\rho}}_i$

$\dot{\boldsymbol{\rho}}_i$ = relative velocity of m_i wrt G

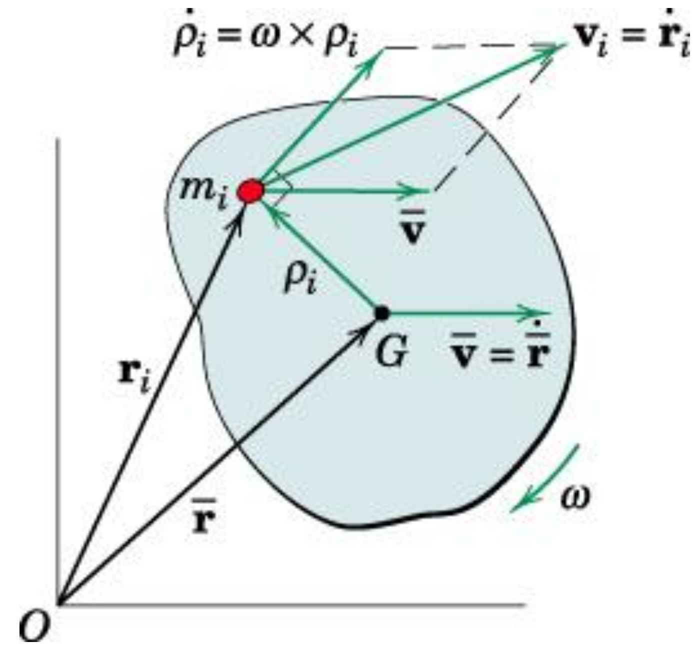
$\dot{\boldsymbol{\rho}}_i = \boldsymbol{\omega} \times \boldsymbol{\rho}_i$ and its magnitude = $\rho_i\,\omega$

Magnitude of $\boldsymbol{\rho}_i \times m_i \dot{\boldsymbol{\rho}}_i = \rho_i^{\,2}\omega m_i$

With $\boldsymbol{\omega} = \omega\mathbf{k}$

$$\mathbf{H}_G = \Sigma \rho_i^{\,2} m_i \omega \mathbf{k} = \bar{I}\omega\mathbf{k}$$

$$\boxed{H_G = \bar{I}\omega}$$



Moment-Angular Momentum Relation:

$$\boxed{\Sigma M_G = \dot{H}_G}$$ and $$\boxed{(H_G)_1 + \int_{t_1}^{t_2} \Sigma M_G \, dt = (H_G)_2}$$

$M\,dt$ = Angular Impulse

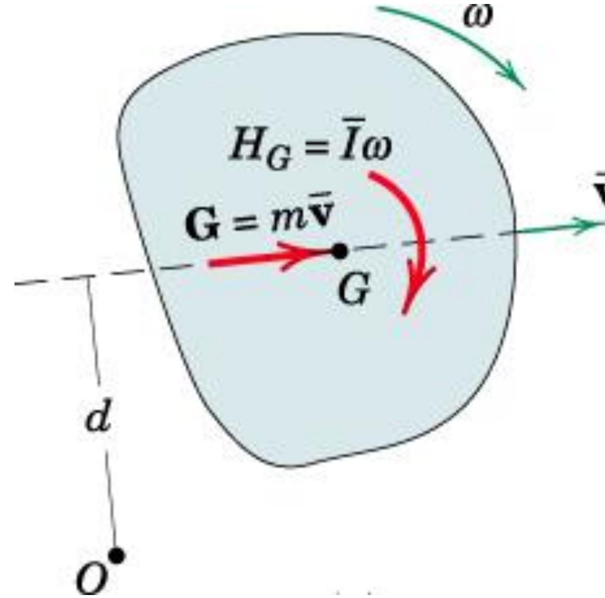
Angular Momentum

•**G** and $\mathbf{H}_G$ have **vector** properties **analogous** to those of the **resultant force** and **couple**

→ Angular momentum @ any point *O*:

$$\boxed{H_O = \bar{I}\omega + m\bar{v}d}$$

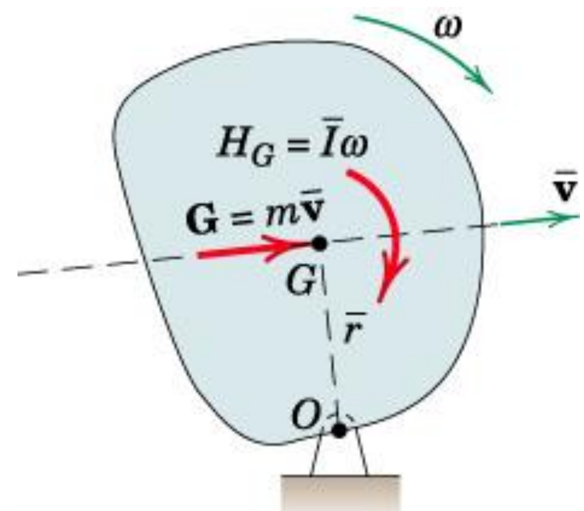
→ Applicable at any particular instant of time @ *O*, which may be a fixed or moving point on or off the body



When the body rotates @ a fixed point *O* on the body or body extended, $v = \bar{r}\omega$ and $d = \bar{r}$

→ $H_O = (\bar{I}\omega + m\bar{r}^2\omega)$

$$\bar{I} + m\bar{r}^2 = I_O \qquad \boxed{H_O = I_O\omega}$$



Similarly,

$$\boxed{\Sigma M_O = \dot{H}_O} \quad \text{and} \quad \boxed{(H_O)_1 + \int_{t_1}^{t_2} \Sigma M_O \, dt = (H_O)_2}$$

Rigid Body Kinetics :: Impulse/Momentum

Interconnected Rigid Bodies

For a single rigid body,

$$\Sigma \mathbf{F} = \dot{\mathbf{G}} \qquad \Sigma M_O = \dot{H}_O$$

O :: fixed reference point for the entire system

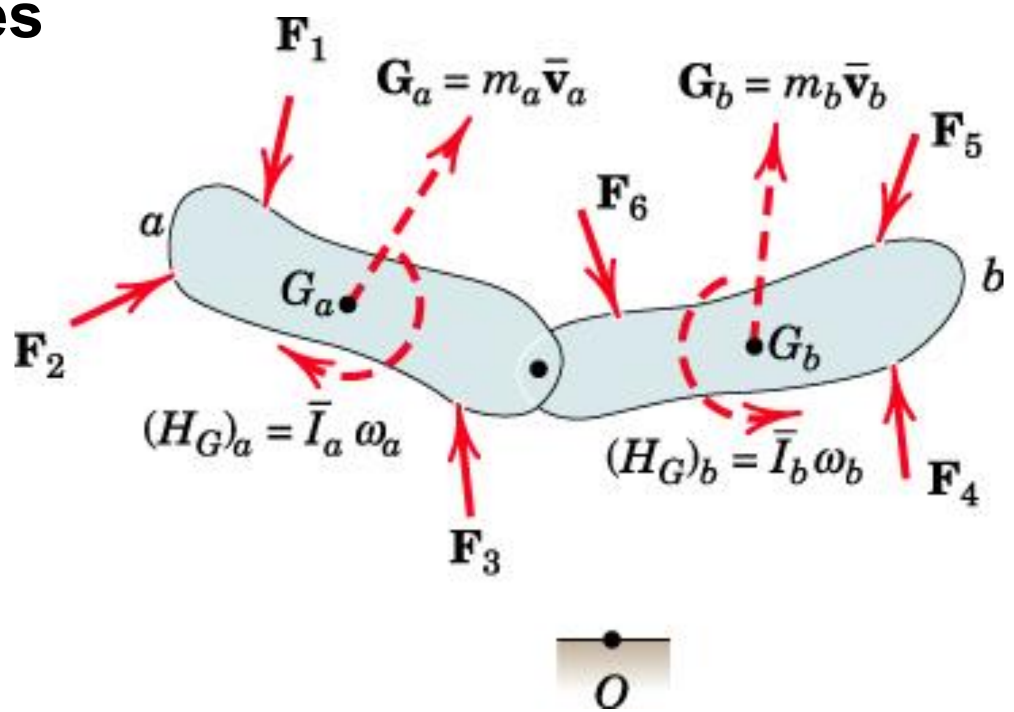
For the system,

$$\Sigma \mathbf{F} = \dot{\mathbf{G}}_a + \dot{\mathbf{G}}_b + \cdots$$

$$\Sigma \mathbf{M}_O = (\dot{\mathbf{H}}_O)_a + (\dot{\mathbf{H}}_O)_b + \cdots$$

Over a finite time interval,

$$\int_{t_1}^{t_2} \Sigma \mathbf{F}\, dt = (\Delta \mathbf{G})_{\text{system}} \qquad \int_{t_1}^{t_2} \Sigma \mathbf{M}_O\, dt = (\Delta \mathbf{H}_O)_{\text{system}}$$



Rigid Body Kinetics :: Impulse/Momentum

Conservation of Momentum

For a single rigid body or a system of interconnected rigid bodies:

$\boxed{\mathbf{G}_1 = \mathbf{G}_2}$ if $\sum \mathbf{F} = \mathbf{0}$ for a given interval of time

→ linear-momentum vector undergoes no change in the absence of a resultant linear impulse.

→For interconnected rigid bodies, linear momentum may change in individual parts during the interval, but there will be no resultant momentum change for the whole system if there is no resultant linear impulse

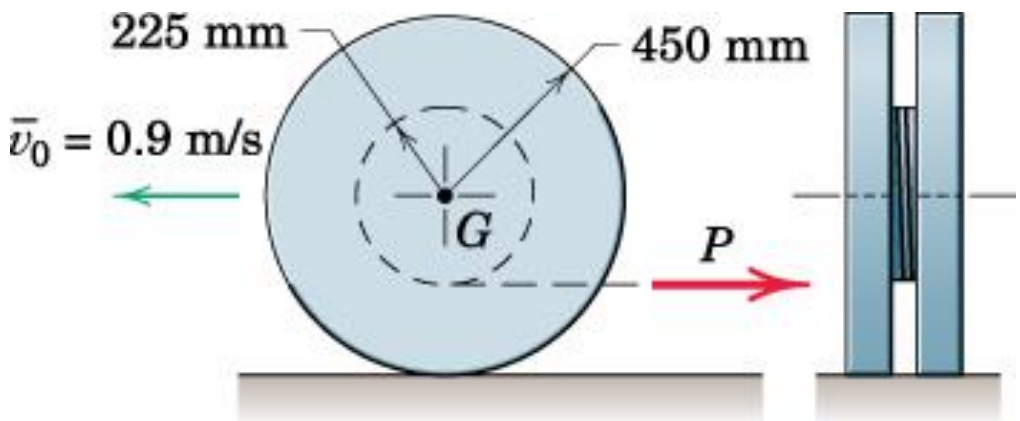
Similarly, if $\sum M_O = 0$ or $\sum M_G = 0$ for a given interval of time

$\boxed{(\mathbf{H}_O)_1 = (\mathbf{H}_O)_2}$ or $\boxed{(\mathbf{H}_G)_1 = (\mathbf{H}_G)_2}$

→angular-momentum either @ a fixed point or @ mass center undergoes no change in the absence of a corresponding resultant angular impulse
→For interconnected rigid bodies, ang momentum may change in individual parts, but no change for the whole system if there is no resultant angular impulse

Example (1) on impulse/momentum

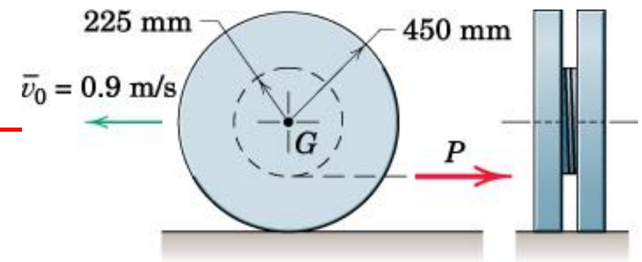
A force *P*, applied to the cable wrapped around the wheel, is increased slowly according to $P = 6.5t$, where *P* is in N and *t* is time in sec after *P* is first applied. At time $t = 0$, the wheel is rolling (without slipping) to the left with a velocity of its center of 0.9 m/s. Wheel has a mass of 60 kg and radius of gyration @ its center is 250 mm. Determine the angular velocity of the wheel 10 seconds after *P* is applied.

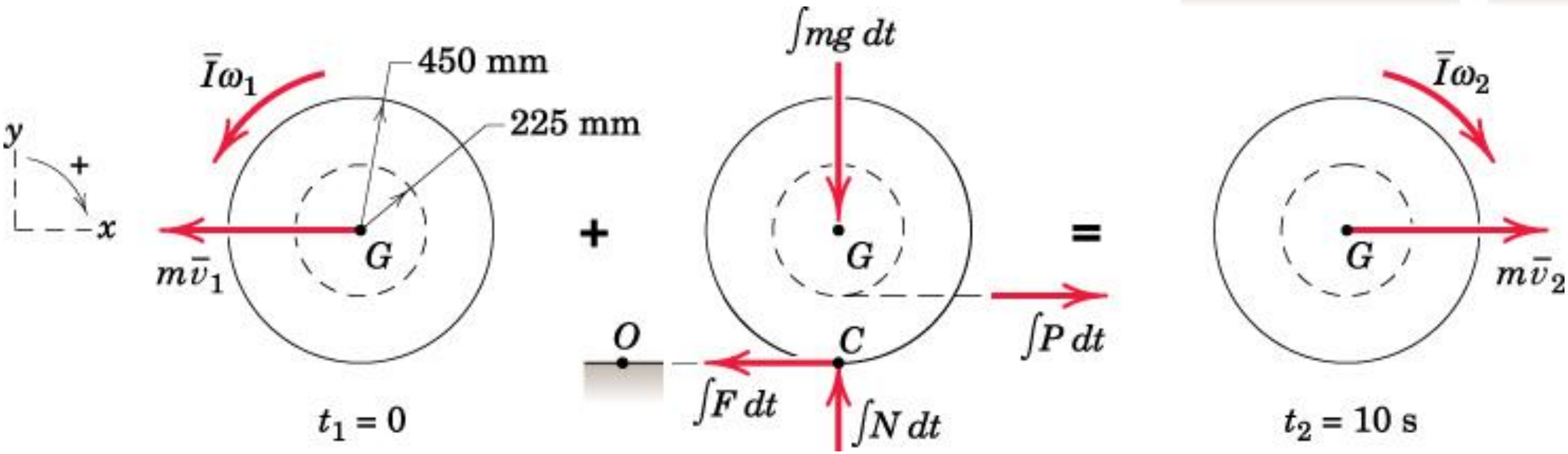


Solution

Draw the impulse-momentum diagrams of the wheel at initial and final stages

Example (1) on impulse/momentum



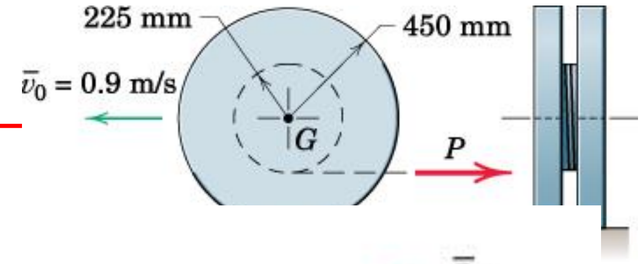


$$\left[(G_x)_1 + \int_{t_1}^{t_2} \Sigma F_x\, dt = (G_x)_2\right]$$

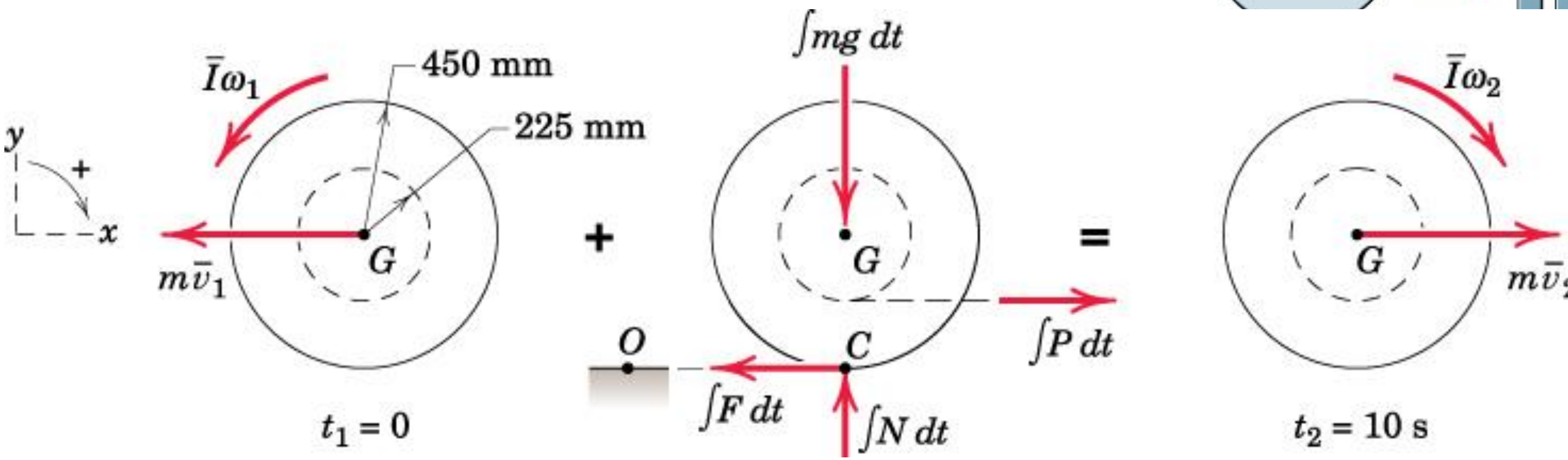
$$60(-0.9) + \int_0^{10} (6.5t - F)dt = 60(0.45\omega_2)$$

Note carefully the signs of the momentum terms. The final linear velocity is assumed in the positive x-direction, so $(G_x)_2$ is positive. The initial linear velocity is negative, so $(G_x)_1$ is negative.

Example (1) on impulse/momentum



Solution



$$\left[(G_x)_1 + \int_{t_1}^{t_2} \Sigma F_x\,dt = (G_x)_2\right] \qquad 60(-0.9) + \int_0^{10}(6.5t - F)dt = 60(0.45\omega_2)$$

$$\left[(H_G)_1 + \int_{t_1}^{t_2} \Sigma M_G\,dt = (H_G)_2\right] \qquad 60(0.25)^2\left(-\frac{0.9}{0.45}\right) + \int_0^{10}[0.45F - 0.225(6.5t)]dt = 60(0.25)^2\omega_2$$

Eliminating F by multiplying 2nd eqn by 1/0.45 and adding the two eqns.
Integrating and solving → ω_2 = 2.6 rad/s (clockwise)